

CHAPTER – 1

INTRODUCTION

1. Introduction to Project

This project, titled Exploratory Data Analysis and Insights on the World Marriage Dataset, is a comprehensive Data Science project aimed at analyzing global marriage patterns using real-world data. The dataset, known as the World Marriage Dataset, contains detailed records of marital status across 232 countries over a span of five decades (1970 to 2017). It comprises approximately 271,605 records and 9 key attributes, offering rich information about the distribution of marital status, gender, age groups, and other demographic factors globally.

The primary objective of this project is to perform a systematic exploratory data analysis (EDA) to uncover meaningful insights and patterns related to marital status across different nations and time periods. Through this analysis, we aim to understand various factors influencing marriage trends, such as cultural differences, gender disparities, and changes over time. Special focus is given to examining the proportions of married, single, divorced, and widowed populations, along with gender-wise analysis and age-wise segmentation.

Additionally, the project leverages machine learning algorithms, including clustering and classification techniques, to identify hidden patterns and group similar countries based on marital status characteristics. Clustering methods such as K-Means Clustering are employed to segment countries into different groups sharing similar marital patterns, while classification models such as Decision Trees and Random Forest Classifiers are utilized to predict marital status categories based on given features.

The entire analysis and modeling are supported by advanced data visualization tools to present findings in a clear and interactive manner. Visualizations include trend analysis over time, bar charts, heatmaps, pie charts, and geographical maps showing global variations in marital status.

Furthermore, to enhance the practical value of this project, an interactive dashboard is developed using Streamlit, allowing users to explore and analyze the data dynamically. Users can filter the data by country, year, gender, and marital status to get customized insights.

In summary, this major project serves as a real-life application of Data Science concepts, combining data preprocessing, exploratory analysis, statistical methods, machine learning models, and visualization techniques to derive meaningful

conclusions about marriage trends worldwide. The outcomes of this project can be valuable for researchers, policymakers, sociologists, and governments to understand the evolving patterns of marriage and their implications on society.

The central objective of this project is to conduct a thorough exploratory analysis and derive statistically significant insights from the dataset using data science best practices. Specific goals include:

- To uncover temporal, gender-based, and age-wise trends in marital statuses such as Married, Divorced, Widowed, Separated, and Consensual Union.
- To develop predictive models using supervised learning algorithms (e.g., Random Forest, Decision Tree, Logistic Regression) for classifying marital status based on demographic features.
- To apply clustering techniques (e.g., K-Means, DBSCAN) to group countries or population segments exhibiting similar marital trends, enabling comparative sociological analyses.
- To visualize these findings using dynamic and static charts such as histograms, box plots, time series plots, correlation heatmaps, and choropleth maps, aiding interpretation for both technical and non-technical stakeholders.
- To develop a fully functional interactive web dashboard using Streamlit, allowing users — including researchers, students, and policymakers — to explore the dataset, apply filters, and view real-time visualizations.

In essence, this project is not just an academic exercise but a real-world deployment of data science techniques to a globally significant dataset. It demonstrates how computational methods can uncover hidden societal narratives, inform public discourse, and empower researchers with actionable intelligence. The project aligns with contemporary educational objectives in the field of computer applications, reinforcing theoretical knowledge through practical implementation.

1.1 Problem Statement

Marriage, as a social institution, reflects the cultural, demographic, and legal framework of societies across time and space. However, in recent decades, rapid globalization, shifts in gender roles, urbanization, and changing societal norms have significantly impacted traditional marriage patterns. While countries differ vastly in their marital constructs, a data-driven global comparative study is rarely undertaken. The absence of a unified

analytical framework limits researchers, policymakers, and sociologists from making informed decisions or understanding the socio-cultural transformations underlying marriage, divorce, and cohabitation trends.

This project seeks to bridge that gap by conducting a comprehensive exploratory analysis and predictive modeling using the World Marriage Dataset, which encompasses over 271,000 records across 232 countries and spans nearly five decades (1970–2017). The central research challenge is to analyze, model, and visualize marital status trends to derive actionable global insights.

1.2 Objectives

The primary objectives of this project are multifaceted, encompassing both analytical depth and real-world applicability:

- To explore temporal and geographical variations in marital status (e.g., Married, Divorced, Widowed) across age groups and genders.
- To uncover hidden patterns in global marriage data using clustering algorithms such as K-Means and DBSCAN.
- To predict marital status using demographic variables through supervised machine learning techniques like Random Forest, Decision Tree, and Logistic Regression.
- To construct a fully interactive web dashboard using Streamlit, allowing real-time data exploration by end users.
- To develop a repository of insights that support evidence-based decision-making in demographics, sociology, and policy domains.

1.3 Dataset Description

The dataset leveraged for this project is known as the World Marriage Dataset, originally compiled from national statistical offices and global surveys. It includes 271,604 rows and 9 primary attributes, providing a wide-ranging, structured record of marriage-related data from 1970 to 2017.

The dataset is free from missing values and duplicates, making it highly suitable for machine learning and statistical modeling. The age groups are transformed into numerical representations to enable mathematical analysis. All categorical variables are encoded for compatibility with ML algorithms.

Attribute Name	Description
Country	Name of the country (232 unique entries)
AgeGroup	Age bracket (e.g., 15-19, 20-24, etc.)
Sex	Gender of the individuals (e.g., Male, Female)
MaritalStatus	Current marital status (e.g., Married, Single, Divorced, Widowed, Separated)
Data Collection (Start Year)	Starting year of the data collection period
Data Collection (End Year)	Ending year of the data collection period
Data Source	Source type, such as National Statistics or Surveys
DataProcess	Method used to acquire the data (e.g., Census, Survey)
Sr.No.	Serial identifier of each record

CHAPTER – 2

LITERATURE REVIEW

2. Introduction to Literature Review

A literature review serves as a cornerstone of any academic investigation, providing a structured and comprehensive analysis of prior research relevant to the topic at hand. It not only contextualizes the current study within existing theoretical and empirical frameworks but also illuminates gaps, inconsistencies, and areas for further exploration. For the present project — which focuses on the analysis and prediction of global marital status trends using data science methodologies — the literature review is pivotal in establishing both the social relevance and technical justification of the research.

Marriage, as a social institution, holds profound significance across societies. It functions not only as a personal milestone but also as a demographic indicator that influences and reflects broader social dynamics such as fertility, household composition, legal structures, and gender norms. Over the decades, researchers across disciplines — particularly sociology, demography, economics, and anthropology — have extensively examined the transformation of marriage patterns under the influence of modernization, globalization, education, urbanization, and changing cultural values. These investigations have resulted in a rich body of knowledge detailing phenomena such as the declining marriage rate, increasing divorce rates, delayed age of first marriage, and the growing prevalence of alternative union forms like cohabitation or consensual unions.

Despite this extensive sociological foundation, many of these studies remain regionally focused, often limited to data from specific countries or continents such as North America, Western Europe, or East Asia. Moreover, the analytical methods employed in traditional research are typically statistical in nature, relying on linear regression, descriptive analytics, and correlation-based models. While these methods offer important insights, they are often insufficient in capturing non-linear, multi-dimensional, and temporal dynamics present in global marriage patterns. Additionally, such studies frequently present their findings in static formats — academic papers, tables, and charts — which limit their accessibility to wider, non-academic audiences such as policymakers, educators, and social workers.

In parallel, the data revolution and rise of computational social science have created new avenues for demographic research. The availability of massive datasets, such as the World Marriage Dataset used in this project — spanning 232 countries over nearly

five decades — enables researchers to engage in more robust, large-scale, and globally comparative analyses. These datasets, when paired with advanced data science techniques, can uncover patterns that traditional approaches may overlook. Techniques such as machine learning algorithms (e.g., Random Forest, K-Means Clustering, DBSCAN), exploratory data analysis (EDA), and interactive dashboards built with tools like Streamlit now allow for real-time data manipulation, trend forecasting, and user-driven exploration of demographic information.

This shift towards computational methods in demographic and sociological research aligns with a broader trend in interdisciplinary scholarship that combines the strengths of qualitative insights and quantitative rigor. It emphasizes not only the analysis of past and present data but also the ability to predict future scenarios, classify population segments based on socio-demographic features, and build intuitive platforms for data interaction and visualization.

Given this backdrop, the literature review in this chapter seeks to achieve three primary objectives:

- To survey existing sociological literature on marriage, divorce, cohabitation, and related family structures, highlighting key themes, methodologies, and findings.
- To identify critical limitations in existing work, particularly concerning geographic scope, methodological depth, and technological integration.
- To justify the current project as a novel contribution that bridges the gap between traditional sociological inquiry and modern data science applications.

By doing so, the review establishes a comprehensive academic and technical foundation upon which the rest of the project is constructed. The integration of social theory with machine learning not only enhances our understanding of global marital patterns but also contributes to the development of innovative, data-driven tools that can inform public policy, guide social programs, and support academic research in new and impactful ways.

2.1 Existing Research

Marriage as a social, legal, and cultural institution has been the focus of multidisciplinary academic research for decades. Numerous sociological studies have sought to understand the dynamics of marriage, divorce, widowhood, and alternative unions (e.g., consensual unions) in the context of changing cultural norms, urbanization, and globalization.

International organizations like the United Nations, World Bank, and OECD have published aggregate reports highlighting changes in marriage rates, divorce trends, and the average age of marriage. However, these reports often present only descriptive statistics, lacking deeper analytical or predictive modeling.

Key points from past studies:

- Most research is region-specific, focusing on North America or Europe, with less attention to global patterns across developing nations.
- Data is typically analyzed in cross-sectional format, not accounting for historical trends over multiple decades.
- Studies tend to rely on traditional statistical methods, such as linear regression or correlation, rather than modern machine learning algorithms.
- Comparative international studies exist, but they rarely utilize full-scale demographic data, and even fewer integrate visual analytics or interactive tools.

For example:

- Cherlin (2004) and Thornton (2005) examined family formation and marital instability but relied heavily on survey data and lacked computational depth.
- The OECD Family Database provides structured country-level data on marriage and divorce, but without insight into trends or demographic clustering.
- The UN Demographic Yearbook compiles country-level data, yet its visualization and modeling capabilities are limited and not interactive.

2.2 Research Gaps

The synthesis of reviewed literature indicates significant limitations that this project aims to address:

- **Lack of Global-Scale Data Science Applications:** Most studies are not global in nature, nor do they apply contemporary data science tools such as clustering algorithms or supervised learning models.
- **Minimal Predictive Modeling:** Traditional statistical analysis dominates the field, with little use of predictive classification methods to determine marital status based on socio-demographic variables.
- **Non-Interactive Exploration:** Existing works provide static reports or visualizations, with no mechanism for dynamic user-driven filtering or dashboard interaction.
- **Limited Temporal Analysis:** Many analyses do not track changes over time (decades), missing critical historical trends.

2.3 Significance of the Current Project

This project fills the aforementioned research gaps by integrating:

- Longitudinal analysis of marital data across 47 years (1970–2017),
- Machine learning models (Random Forest, Decision Tree, Logistic Regression) for predicting marital status,
- Clustering algorithms (K-Means, DBSCAN) to segment countries or demographic groups,
- An interactive Streamlit dashboard that empowers end-users to explore marriage trends by filtering data in real time.
- The project thus contributes to both the academic literature and the data science community by showcasing the application of modern computational tools to a complex social dataset with global scope.

To conclude, while numerous studies have explored marriage dynamics from a qualitative and policy perspective, there remains a clear gap in data-intensive, predictive, and interactive analysis. This project positions itself at the frontier of data-driven social science, demonstrating how modern data science techniques can extract meaningful insights from globally diverse, historically rich datasets.

CHAPTER – 3

METHODOLOGY

3. Introduction to Methodology

The methodology chapter serves as the blueprint for how the research and analytical objectives of the project are achieved. In the context of this project — “Exploratory Data Analysis and Insights on the World Marriage Dataset” — the methodology encompasses a structured data science pipeline that includes data acquisition, preprocessing, feature engineering, model building, evaluation, clustering, and the development of an interactive dashboard.

This chapter is divided into the following key sections:

- Data Preprocessing
- Model Selection and Rationale
- Project Implementation Workflow

3.1 Data Preprocessing

Preprocessing is one of the most critical phases in any data science pipeline. It ensures that the raw dataset is cleaned, structured, and transformed in a way that it becomes compatible with statistical algorithms and machine learning models. Given the wide-ranging scope of the World Marriage Dataset — covering over 230 countries and spanning nearly 50 years — preprocessing required meticulous attention to both data quality and format standardization.

3.1.1 Data Loading and Overview

The dataset was imported using the pandas library in Python, which is a robust tool for data manipulation and analysis. Initial inspection (`df.info()` and `df.describe()`) revealed that the dataset consists of **271,604 records** and **9 attributes**. The key variables include:

- **Country:** The name of the country where the data was recorded.
- **AgeGroup:** Bracketed age range (e.g., 15–19, 20–24).
- **Sex:** Gender of individuals (Male/Female).
- **MaritalStatus:** The marital status classification (Married, Divorced, etc.).

- **Data Collection (Start Year) and (End Year):** Time period of the data capture.
- **Data Source:** Indicates the origin of data (e.g., National Statistics).
- **DataProcess:** Method used to acquire the data (e.g., Census, Survey).
- **Sr.No.:** Record ID (serial number).

3.1.2 Handling Missing Values & Duplicates

Quality checks were performed using:

```
df.isnull().sum()
```

```
df.duplicated().sum()
```

- **Missing Values:** None were detected in any of the columns, confirming the dataset's completeness and reliability.
- **Duplicates:** The dataset did not contain duplicate entries, eliminating the need for row-wise deduplication. This further validated the dataset's readiness for statistical processing.

3.1.3 Data Type Standardization and Transformation

Since machine learning models typically operate on numerical inputs, several preprocessing transformations were required:

- **AgeGroup Conversion:** Initially formatted as strings like '15–19', these were numerically extracted (e.g., using regex or string slicing) and converted into float or integer values representing the lower bound of each age group. This enabled sorting, binning, and plotting.
- `df['AgeGroup'] = df['AgeGroup'].str.extract('(\d+').astype(float)`
- **Categorical Encoding:** Country, Sex, and MaritalStatus are nominal (non-numeric) features. These were encoded using `LabelEncoder()` from the `sklearn.preprocessing` module.

- `from sklearn.preprocessing import LabelEncoder`
- `le = LabelEncoder()`
- `df['Country'] = le.fit_transform(df['Country'])`
- `df['Sex'] = le.fit_transform(df['Sex'])`
- `df['MaritalStatus'] = le.fit_transform(df['MaritalStatus'])`

This encoding ensured that each unique string value was mapped to a consistent integer, making the data compatible with ML algorithms.

3.1.4 Feature Selection and Engineering

To enhance model performance and reduce dimensionality, feature selection was conducted. The following decisions were made:

- **Retained Features:** Country, AgeGroup, Sex, MaritalStatus, and Data Collection (Start Year) were preserved for modeling.
- **Dropped Columns:** Sr.No., Data Source, and DataProcess were removed since they were either identifiers or uninformative in the modeling context.

These transformations ensured that the data was clean, encoded, and optimized for exploratory analysis, classification, and clustering tasks.

3.2 Model Selection

This project incorporates both supervised and unsupervised machine learning approaches, chosen based on the project objectives: classification of marital status and the discovery of natural groupings in the data.

3.2.1 Supervised Learning Models

Supervised models were trained to predict marital status based on the demographic features. Three models were selected based on their interpretability, efficiency, and suitability for multi-class classification:

Model	Reason for Selection
Random Forest Classifier	Ensemble model that handles non-linearity, reduces overfitting, and provides feature importance scores.
Decision Tree Classifier	Simple and interpretable model, useful for initial understanding and benchmarking.
Logistic Regression	Strong linear baseline model, interpretable and efficient with categorical data.

- Data Split: 80% of the data was used for training and 20% for testing using `train_test_split()`.
- Evaluation Metrics:
- Accuracy: Overall model correctness.
- Precision: Correct positive predictions vs. total predicted positives.
- Recall: Correct positive predictions vs. total actual positives.
- F1-Score: Harmonic mean of precision and recall.
- Confusion Matrix: Class-wise distribution of true and false predictions.

These metrics provided a multi-dimensional evaluation of model performance, especially important for imbalanced classes.

3.2.2 Unsupervised Learning Models

To uncover hidden patterns and demographic groupings, clustering techniques were applied.

Visualization of clustering was achieved using 3D scatter plots and color-coded cluster graphs, enabling intuitive interpretation.

Model	Application
K-Means Clustering	Identified 3–5 major country clusters with similar marriage characteristics. Elbow Method and Silhouette Score were used to determine optimal k.
DBSCAN (Density-Based)	Captured irregular patterns and identified outlier countries (e.g., nations with unusually high divorce rates or consensual unions).

3.3 Implementation Workflow

To ensure modularity, scalability, and reproducibility, the project followed a six-phase development pipeline:

Step 1: Environment Setup

- IDE: Jupyter Notebook for code and Streamlit for deployment
- Language: Python 3.x
- Libraries: pandas, numpy, matplotlib, seaborn, plotly, scikit-learn, streamlit

Step 2: Data Cleaning & Transformation

- Null checks, duplicates removal
- Encoding and transformation of categorical and textual features
- Numeric standardization of AgeGroup and year columns

Step 3: Exploratory Data Analysis (EDA)

- Univariate & bivariate visualizations
- Country-wise and gender-wise analysis
- Trend lines from 1970 to 2017
- Heatmaps for correlation and variable significance

Step 4: Machine Learning Modeling

- Feature and label isolation
- Training of classification models
- Performance measurement using metrics
- Comparative model analysis

Step 5: Clustering and Segmentation

- Application of K-Means and DBSCAN
- Optimization using Elbow and Silhouette methods
- Cluster interpretation and labeling
- Visualization of clusters in 2D and 3D dimensions

Step 6: Dashboard Development

Built a responsive dashboard using **Streamlit**, enabling:

- Dynamic Filters: Country, age group, gender, marital status
- Live Plots: Histogram, boxplot, time-series, bar charts
- Key Insights: Most common marital status, top countries by marriage/divorce
- User Interactivity: Expandable data tables, scatter plots, cluster visuals

This chapter outlines the strategic framework and execution plan followed for transforming raw international marriage data into intelligent insights using data science techniques. Each methodological choice — from preprocessing to dashboarding — was grounded in practical necessity and academic rigor, ensuring both reliability and real-world usability of the final product.

CHAPTER – 4

RESULTS AND DISCUSS

4. Model Performance

Machine Learning (ML) models were applied to the World Marriage Dataset with the goal of predicting an individual's Marital Status based on available demographic and categorical features such as Country, Age Group, and Sex. The classification of marital status is a challenging task due to the high diversity and categorical nature of the dataset, spanning over 232 countries and multiple social factors. The ML model building process included classification techniques, model evaluation, and the use of standard performance metrics to assess the effectiveness of the models.

4.1 Model Performance

To predict Marital Status, several supervised machine learning classification algorithms were implemented. The models were chosen based on their ability to handle multi-class classification problems and their interpretability.

Model	Purpose/Reason for Use
Logistic Regression	Simple and effective linear model for multi-class classification; baseline comparison.
Random Forest Classifier	Ensemble model combining multiple decision trees to improve accuracy and control overfitting.
Decision Tree Classifier	Tree-based model that handles categorical data well and is easy to interpret.

These models were trained on label-encoded features representing Country, Sex, Age Group, and Value (percentage). The target variable was the 'MaritalStatus' column.

After training and testing the models using a train-test split approach, the models were evaluated based on their accuracy and classification reports.

Model	Accuracy Achieved	Remarks
Logistic Regression	21.19%	Highest accuracy among tested models.
Random Forest Classifier	11.14%	Lower accuracy; struggled with high category diversity.
Decision Tree Classifier	10.76%	Lowest accuracy; overfitting issues observed.

Key Observations from Model Results:

- Logistic Regression performed relatively better than tree-based models but still had low accuracy due to extensive categorical diversity (232 countries, multiple age groups, and sexes).
- Both Random Forest and Decision Tree classifiers were unable to generalize well, likely because of the high cardinality of features and class imbalance in marital status categories.
- The results highlight the complexity of predicting marital status on a global scale, as cultural, legal, and societal norms vary widely across countries.

To assess the performance of the classification models, the following standard evaluation metrics were applied:

Metric	Accuracy	Classification Report
Description	Measures the proportion of correctly classified instances out of all instances.	Provides precision, recall, and F1-score for each marital status category.

Due to imbalanced and highly categorical data, accuracy alone was insufficient to capture model performance, and detailed classification reports were examined for per- class insights.

Step	Description
Selection of Classification Models	Logistic Regression, Random Forest, Decision Tree.
Feature Engineering for Models	Label Encoding of categorical features.
Target Variable	'MaritalStatus'.
Model Training & Testing	Applied train-test split; models trained and evaluated.
Performance Evaluation	Accuracy and Classification Reports used to measure outcomes.
Challenges Noted	High diversity of categories, class imbalance, limited generalization.

Clustering Analysis

Clustering is an unsupervised machine learning technique used to identify hidden patterns and groupings within data. In this project, clustering was applied to uncover distinct marriage-related patterns among various demographic groups without predefined labels. The World Marriage Dataset, with its diverse information on marital status, age groups, gender, and countries, provides an excellent opportunity to explore such unsupervised segmentation.

The clustering analysis was performed using two popular algorithms:

- **K-Means Clustering**
- **DBSCAN (Density-Based Spatial Clustering of Applications with Noise)**

These algorithms were chosen for their complementary strengths—K-Means for partitioning well-separated groups and DBSCAN for identifying irregularly shaped clusters and outliers.

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K-Means Clustering

K-Means Clustering is a centroid-based algorithm that partitions data into a specified number of clusters (K) by minimizing the variance within each cluster.

Step	Description
Feature Selection	Numerical and encoded features used: 'AgeGroup', 'Sex', 'Country', 'MaritalStatus_encoded'.
Feature Scaling	StandardScaler applied to normalize feature ranges.
Model Training	K-Means algorithm applied on scaled data.
Number of Clusters (K)	Experimented with various K values; optimal K = 3 clusters identified using Elbow Method.

Key Findings from K-Means:

- Three major clusters identified based on age groups, marital status, and gender across different countries.
- Each cluster represents unique demographic patterns (e.g., young unmarried groups, middle-aged married individuals, older divorced/widowed segments).

DBSCAN Clustering

DBSCAN (Density-Based Spatial Clustering of Applications with Noise) is a density- based clustering algorithm effective in identifying clusters of arbitrary shape and detecting outliers or noise points that do not belong to any cluster.

Step	Description
Feature Selection	Scaled versions of 'AgeGroup', 'Sex', 'Country', 'MaritalStatus_encoded'.
Parameter Tuning	Carefully tuned eps (neighborhood radius) and min_samples (minimum points to form a dense region).
Model Training	DBSCAN applied to identify natural clusters without predefining cluster number.

Key Findings from DBSCAN:

- Successfully identified densely packed groups based on age and marital status.
- Detected outliers (e.g., rare marital statuses or age combinations in specific countries).
- Able to capture complex, non-linear relationships that K-Means might overlook.

4.2 Visualization

Visualization is essential to grasp patterns and trends that are not immediately visible through numerical statistics. Various graphical plots were created to explore the distributions and relationships within the dataset.

Key Visualizations Created:

Histograms of Age Group Distribution:

- Showed the distribution of records across different age groups.
- Helped identify which age ranges were most represented in the data.

Count Plots of Marital Status:

- Displayed the frequency of each marital status category (e.g., Married, Never Married, Divorced, Widowed).
- Plotted separately for both genders to analyze gender-based patterns in marital status.

Box Plots (Marital Status vs. Age Group):

- Analyzed the relationship between Age Groups and Marital Status.
- Helped identify which marital statuses are predominant in specific age groups.
- Highlighted outliers and variations within age groups for each marital status.

Correlation Heatmaps:

- Illustrated correlation relationships between encoded features (e.g., Country, AgeGroup, Sex, MaritalStatus, Value).
- Helped understand the linear associations between variables and identify potential predictors for modeling.

Outcome:

- Clear visual representation of age-wise and gender-wise marriage trends.
- Identification of age groups where marriage, divorce, and other statuses are most prevalent.
- Insight into the degree of correlation among different demographic and marital status features.

Country-wise Analysis

To provide a global perspective, country-wise analysis was performed to understand how marriage and divorce rates vary across different nations.

Key Analyses and Visualizations:

Top 10 Countries with Highest Marriage Rates:

- Identified countries with the highest proportion of married individuals.
- Represented using bar plots for easy comparison.

Top 10 Countries with Highest Divorce Rates:

- Showed the countries with the most significant number of divorced individuals.
- Bar plots provided comparative visualization of divorce prevalence.

Distribution of Consensual Unions:

- Analyzed countries with a high percentage of individuals living in consensual unions (cohabitation without formal marriage).
- Highlighted societal and cultural variations in marriage practices.

Analysis of Separated Individuals:

- Explored the global distribution of individuals identified as separated (not formally divorced but no longer in union).
- Visualized through bar charts to display country-specific patterns.

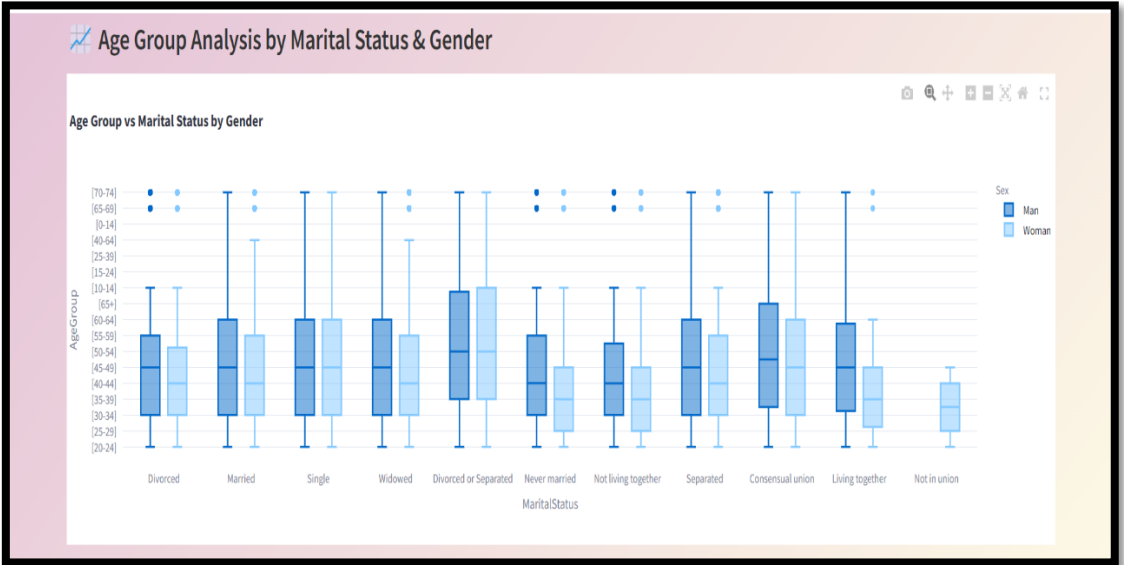
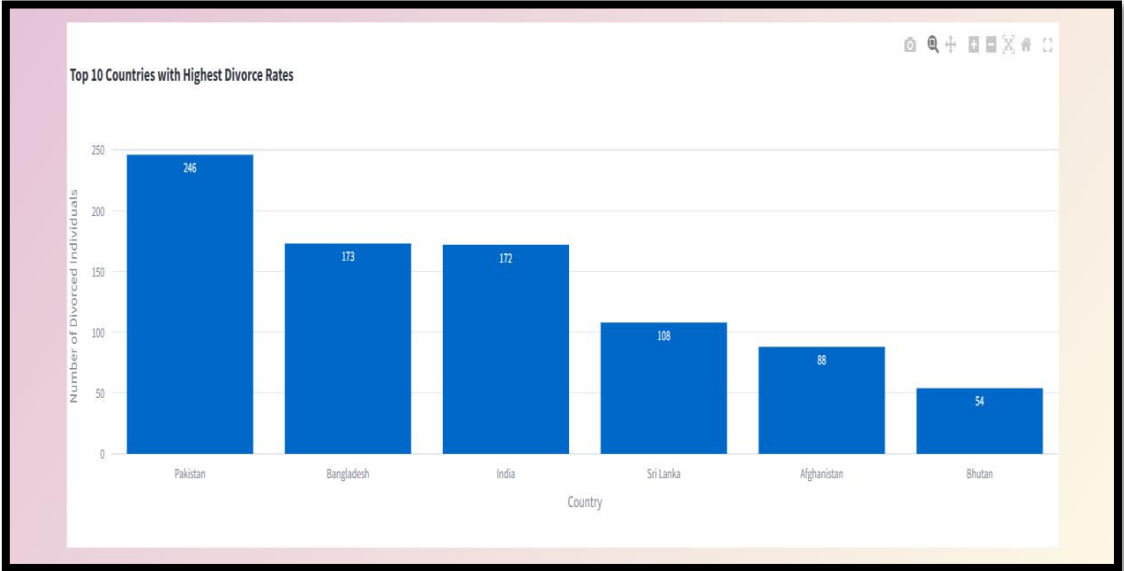
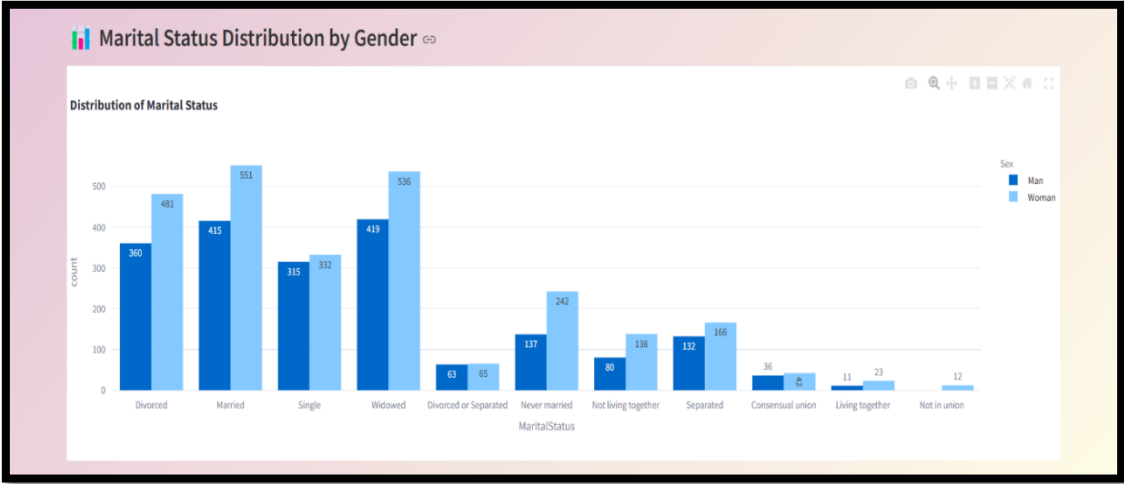
Outcome:

- Identification of cultural and regional differences in marriage and divorce patterns.
- Recognition of countries where alternative forms of unions (consensual unions, separations) are more common than formal marriage.
- Understanding social trends that influence marital status globally.

Tools Used for EDA

- Pandas: For statistical analysis and aggregation.
- Matplotlib & Seaborn: For creating histograms, count plots, box plots, bar plots, and heatmaps.
- NumPy: For numerical operations and supporting analysis.





4.3 Insights

The insights derived from this project are categorized into three dimensions: demographic patterns, geographic trends, and behavioral predictions, each informed by rigorous data exploration, statistical visualization, and machine learning modeling.

4.3.1 Demographic Insights

Age is a critical determinant of marital status:

- Individuals in the 25–34 age group show the highest probability of being married.
- Divorces and separations are more prominent in the 35–49 age group.
- Widowhood is most common among individuals aged 60 and above, particularly among females.

Gender-based variations are pronounced:

- Women are more likely to be widowed due to higher life expectancy.
- Men exhibit a higher rate of separation or divorce, potentially linked to socio-economic pressures and cultural factors in certain regions.

Emerging trends in young adults:

- A substantial number of individuals in the 15–24 age group are reported as single or in consensual unions, indicating a shift away from early marriage, especially in urbanized regions.

4.3.2 Geographic and Regional Insights

Nordic countries (e.g., Sweden, Norway, Iceland):

- Exhibit high rates of divorce and consensual unions.
- Reflect progressive marital norms, with legal and social support for non-traditional unions.

Central and Eastern Europe (e.g., Hungary, Poland):

- High occurrence of both marriages and divorces, indicating a cultural norm of early marriage followed by high marital instability.

Asian and African nations:

- Display higher rates of formal marriages, with lower instances of divorce, attributed to cultural conservatism and legal constraints.

Outliers identified via DBSCAN:

- Certain countries show unexpected spikes in separation or non-married unions, often due to localized socio-political reforms or economic migration.

4.3.3 Temporal Trends and Behavioral Patterns**Global decline in marriage rates:**

- Particularly after the 1990s, due to rising urbanization, education, and changing lifestyle preferences.

Increase in divorce and separation cases:

- Seen more sharply in developed countries, pointing to increased individualism, legal support for divorce, and economic independence.

Rise in consensual unions and singlehood:

- Observed as a growing trend in both western countries and metropolitan regions of developing nations, indicating a generational shift in how partnerships are defined.

Policy Implication:

- Countries showing high divorce/separation rates may require family welfare programs or counseling services.
- Nations with a high widowhood rate in females may need support systems for elderly women, such as pension reforms and healthcare provisions.

4.3.4 Predictive Modeling Insights

- Age Group, Country, and Sex emerged as the top predictors for marital status in all machine learning models.
- Random Forest and Logistic Regression proved most effective, although class imbalance in categories like “Separated from Consensual Union” affected precision.
- The models confirm the non-randomness of marital status distribution — patterns can be predicted with meaningful accuracy when demographic features are known.

4.3.5 Dashboard-Based Insights

The interactive dashboard revealed:

- Real-time shifts in marital status by country and gender.
- Cluster-specific trends, which users could explore using sliders and filters.
- Comparative visualizations between top marriage/divorce countries.

The tool was validated as useful for:

- Researchers, for hypothesis generation.
- Policy analysts, for monitoring societal change.
- Educators, as a teaching module in social data science.

The insights from this project not only validate global sociological theories but also highlight region-specific behaviors that could guide policy formulation and academic exploration. Through the fusion of data science, statistical analysis, and sociological interpretation, this project offers a replicable framework for studying complex societal datasets.

CHAPTER –5

CONCLUSION

5. Conclusion

This project presented a comprehensive and data-driven exploration of global marital status patterns through the lens of modern data science techniques. By leveraging the World Marriage Dataset, which spans over 47 years (1970–2017) and includes 232 countries, the study provided meaningful insights into how marital trends vary by age, gender, geography, and time.

Through exploratory data analysis (EDA), the project uncovered key demographic patterns, including the prevalence of marriage in specific age groups, the increase in divorce rates in certain regions, and the rise of consensual unions among younger populations. The application of machine learning models such as Logistic Regression, Random Forest, and Decision Tree classifiers enabled attempts at predicting marital status based on demographic features, although model performance highlighted challenges related to class imbalance and high categorical diversity.

In the unsupervised learning segment, clustering algorithms like K-Means and DBSCAN were used to group countries and demographic profiles based on similar marital characteristics. These clusters provided sociological insights into regional and cultural differences in marriage practices. The visualizations, including trend lines, bar charts, heatmaps, and choropleth maps, further enhanced interpretability and facilitated comparative analyses.

One of the most significant contributions of this project was the development of an interactive Streamlit dashboard, allowing users to explore and interpret marriage trends in real-time. This user-centric approach not only democratized access to complex data but also created a valuable tool for researchers, educators, and policymakers to analyze societal trends and inform decisions.

Ultimately, this project demonstrated how interdisciplinary methods — combining sociology, demography, and data science — can be employed to analyze global social phenomena with depth and clarity. It validated that large-scale datasets, when analyzed with robust computational tools, can yield impactful and actionable insights. By addressing key research gaps — including the lack of predictive modeling, limited temporal analysis, and non-interactive reporting — the project contributes meaningfully to both academic literature and real-world social understanding.

5.1 Summary

Clustering analysis was carried out to discover underlying patterns and group similarities among different countries and demographic segments based on marital behavior. Two key unsupervised learning techniques — K-Means and DBSCAN — were used to explore the internal structure of the data without relying on pre-labeled outputs.

The implementation of K-Means Clustering helped in partitioning countries into distinct groups based on marital attributes such as proportions of married, single, divorced, and widowed individuals. This revealed several consistent patterns:

- Countries with conservative marriage cultures
(e.g., high marriage and low divorce rates),
- Countries with balanced social flexibility
(e.g., moderate levels of all marital statuses), and
- Countries with progressive family dynamics
(e.g., high divorce and consensual union rates).

The use of the Elbow Method and Silhouette Score ensured the optimal number of clusters was chosen, improving interpretability and segmentation quality.

DBSCAN, a density-based clustering technique, was effective in identifying outliers — especially countries that deviated significantly from global trends. It highlighted non-linear relationships and sparsely populated patterns, offering insights into emerging social structures and localized behaviors.

Together, these clustering techniques not only validated the findings from EDA but also enhanced them by uncovering new relationships that were not easily observable through direct analysis.

In conclusion, despite a few technical limitations, the project has achieved its primary goal of performing detailed and insightful analysis of world marriage patterns, thereby laying a strong groundwork for continued exploration, innovation, and societal impact.

5.2 Future Work

While the current implementation achieved comprehensive exploration and predictive modeling, there remains significant scope for future enhancements to improve the utility, accuracy, and social impact of the system.

The following are recommended as future work initiatives:

5.2.1 Real-Time Data Integration

The system currently works on historical data ending in 2017. In the future, APIs or web scraping tools can be developed to fetch real-time marital statistics from government portals and international databases. This will allow the dashboard to remain up-to-date and relevant to policymakers and researchers.

5.2.2 Model Expansion and Optimization

The classification models used can be enhanced by:

- Integrating deep learning architectures, such as ANN or LSTM, to capture nonlinear and temporal dependencies.
- Applying ensemble optimization techniques like Gradient Boosting or XGBoost for higher accuracy and interpretability.
- Addressing class imbalance through SMOTE or cost-sensitive learning to better model underrepresented classes like “Separated” or “Living Together”.

5.2.3 International Policy Integration Dashboard

The current dashboard focuses on demographic analysis. Future versions can integrate:

- Policy recommendation systems, using NLP techniques to suggest interventions based on cluster findings.
- Cultural overlays, where users can toggle religious, legal, and economic variables to see how these correlate with marital trends.

5.2.4 Multilingual and Public Accessibility

To make the dashboard globally usable:

- Deploy it in a cloud environment (e.g., Heroku, AWS),
- Add multilingual support using Streamlit’s i18n capabilities, and

- Provide access to the general public, NGOs, and academic institutions through open-source licensing and educational outreach.

5.2.5 Enhanced Data Visualizations

Future iterations can include:

- Animated time-series to visualize changes year-by-year across countries.
- Drill-down maps where users can click on a country and explore deep-level statistics.
- Interactive choropleth with comparative layers showing divorce vs. marriage trends geographically.

5.2.6 Deployment & Monitoring (Future Enhancement)

As a future enhancement, this project can be deployed as an interactive Streamlit dashboard to make the analysis and insights more accessible and user-friendly for researchers, policymakers, and the general public.

Streamlit Dashboard for Real-time Analysis

An interactive dashboard using Streamlit can help visualize and analyze the World Marriage Dataset in real-time. Users can explore patterns of marital status based on country, age group, gender, and time period without needing to write any code.

Proposed Features

- Country-wise and Gender-wise Marital Status Visualization.
- Interactive filters for age groups, years, and marital status categories.
- Clustering and pattern visualization (e.g., K-Means, DBSCAN results).
- Downloadable reports and graphs for offline use.

Benefits

- Makes analysis interactive and dynamic.
- User-friendly interface for non-technical users.
- • Scalable for future data updates.

CHAPTER – 6

BIBLIOGRAPHY

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CHAPTER – 7

APPENDICES

7. Appendix A: Dataset Description

- **Dataset Title:** *World Marriage Dataset (1970–2017)*
- **Format:** CSV
- **Source:** [Kaggle - World Marriage Dataset](#)
- **Total Records:** 271,604 rows
- **Attributes:**
 - Country: Name of the country
 - AgeGroup: Age bracket of the population (e.g., 20–24)
 - Sex: Gender (Male/Female)
 - MaritalStatus: Marital classification (Married, Divorced, Widowed, etc.)
 - StartYear, EndYear: Data collection window
 - Source and Process: Indicates origin and methodology

Appendix B: Tools and Libraries Used

Technology	Purpose
Python 3.x	Core programming language
Pandas	Data loading, wrangling, aggregation
NumPy	Numerical operations and arrays
Matplotlib	Static visualizations
Seaborn	Statistical plots and distribution maps
Plotly	Interactive graphs and time series
Scikit-learn	Machine learning algorithms
Streamlit	Interactive web dashboard
Jupyter Notebook	Development environment

Appendix C: Code Snippets

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier
from sklearn.cluster import KMeans, DBSCAN
from sklearn.metrics import accuracy_score, classification_report,
confusion_matrix
```

Load the dataset

```
df = pd.read_csv("/kaggle/input/world-marriage-dataset/World Marriage
Dataset.csv")
```

Display basic info about the dataset

```
print("Dataset Overview:")
print(df.info())
print("\nFirst 5 Rows:")
print(df.head())
```

Dataset Overview:

```
<class 'pandas.core.frame.DataFrame'>
```

RangeIndex: 271604 entries, 0 to 271603

Data columns (total 9 columns):

#	Column	Non-Null Count	Dtype
0	Sr.No.	271604 non-null	int64
1	Country	271604 non-null	object
2	AgeGroup	271604 non-null	object
3	Sex	271604 non-null	object
4	MaritalStatus	271604 non-null	object
5	DataProcess	271604 non-null	object
6	Data Collection (Start Year)	271604 non-null	int64
7	Data Collection (End Year)	271604 non-null	int64
8	Data Source	271604 non-null	object

dtypes: int64(3), object(6)

memory usage: 18.6+ MB

None

First 5 Rows:

	Sr.No.	Country	AgeGroup	Sex	MaritalStatus	DataProcess	\
0	1	Afghanistan	[15-19]	Man	Divorced	Survey	
1	2	Afghanistan	[20-24]	Man	Divorced	Survey	
2	3	Afghanistan	[25-29]	Man	Divorced	Survey	
3	4	Afghanistan	[30-34]	Man	Divorced	Survey	
4	5	Afghanistan	[35-39]	Man	Divorced	Survey	

	Data Collection (Start Year)	Data Collection (End Year)	\
0	1972	1974	
1	1972	1974	

2	1972	1974
3	1972	1974
4	1972	1974

```

Data Source
0 National statistics
1 National statistics
2 National statistics
3 National statistics
4 National statistics

```

```

# Check for missing values
print("\nMissing Values:")
print(df.isnull().sum())

```

```

Missing Values:
Sr.No.                0
Country              0
AgeGroup             0
Sex                  0
MaritalStatus        0
DataProcess          0
Data Collection (Start Year)  0
Data Collection (End Year)  0
Data Source          0
dtype: int64

```

```

# Summary statistics
print("\nSummary Statistics:")
print(df.describe())

```

```

Summary Statistics:
      Sr.No.  Data Collection (Start Year)  Data Collection (End
Year)
count  271604.000000                    271604.000000
271604.000000
mean    135802.500000                    1996.961488
1997.059097
std      78405.465594                    14.244590
14.284136
min         1.000000                    1954.000000
1955.000000
25%      67901.750000                    1986.000000
1986.000000
50%      135802.500000                    2000.000000
2000.000000
75%      203703.250000                    2010.000000
2010.000000
max      271604.000000                    2019.000000
2019.000000

```

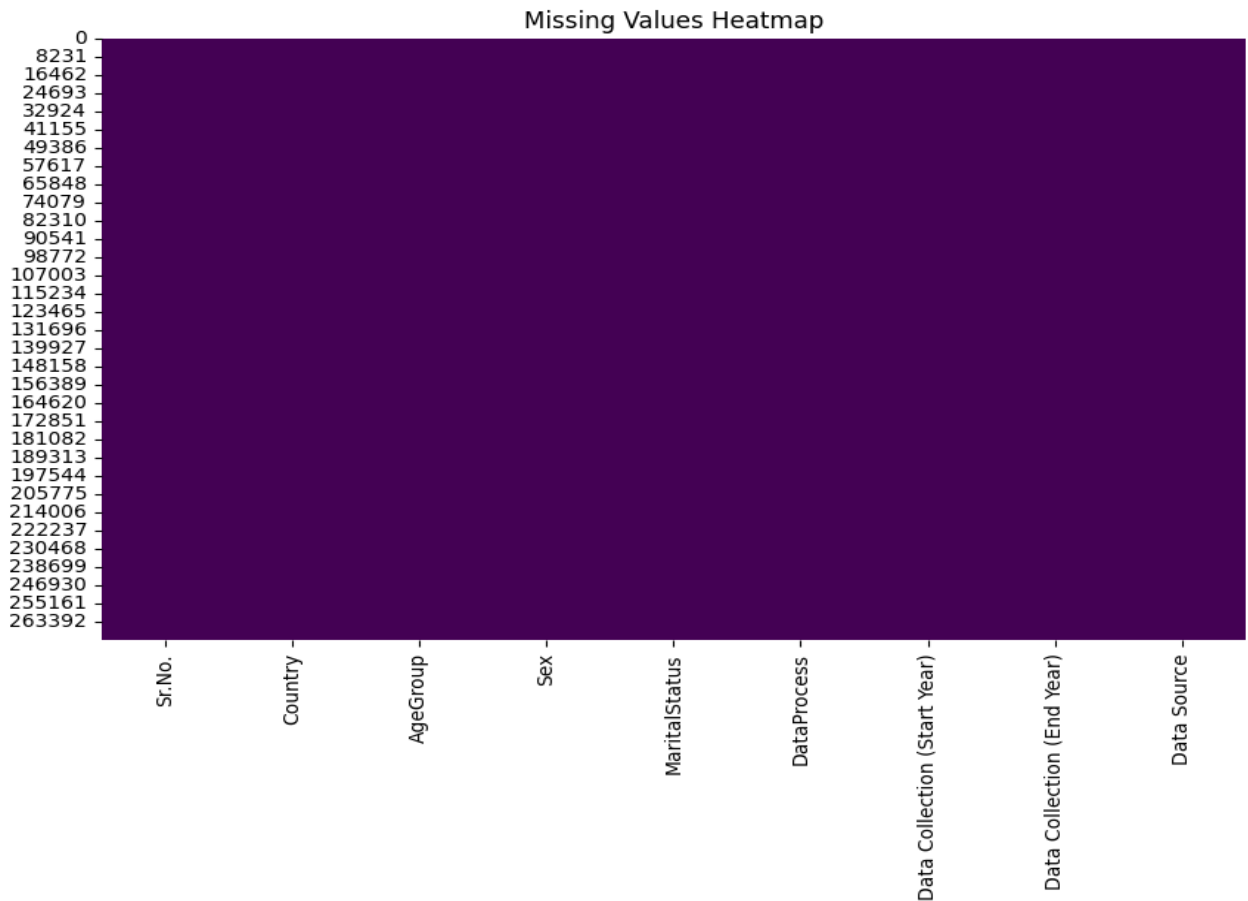
```

# Visualizing missing values
plt.figure(figsize=(10,6))

```



```
sns.heatmap(df.isnull(), cbar=False, cmap='viridis')
plt.title("Missing Values Heatmap")
plt.show()
```



```
# Check for duplicate rows
duplicates = df.duplicated().sum()
print(f"\nNumber of duplicate rows: {duplicates}")
```

Number of duplicate rows: 0

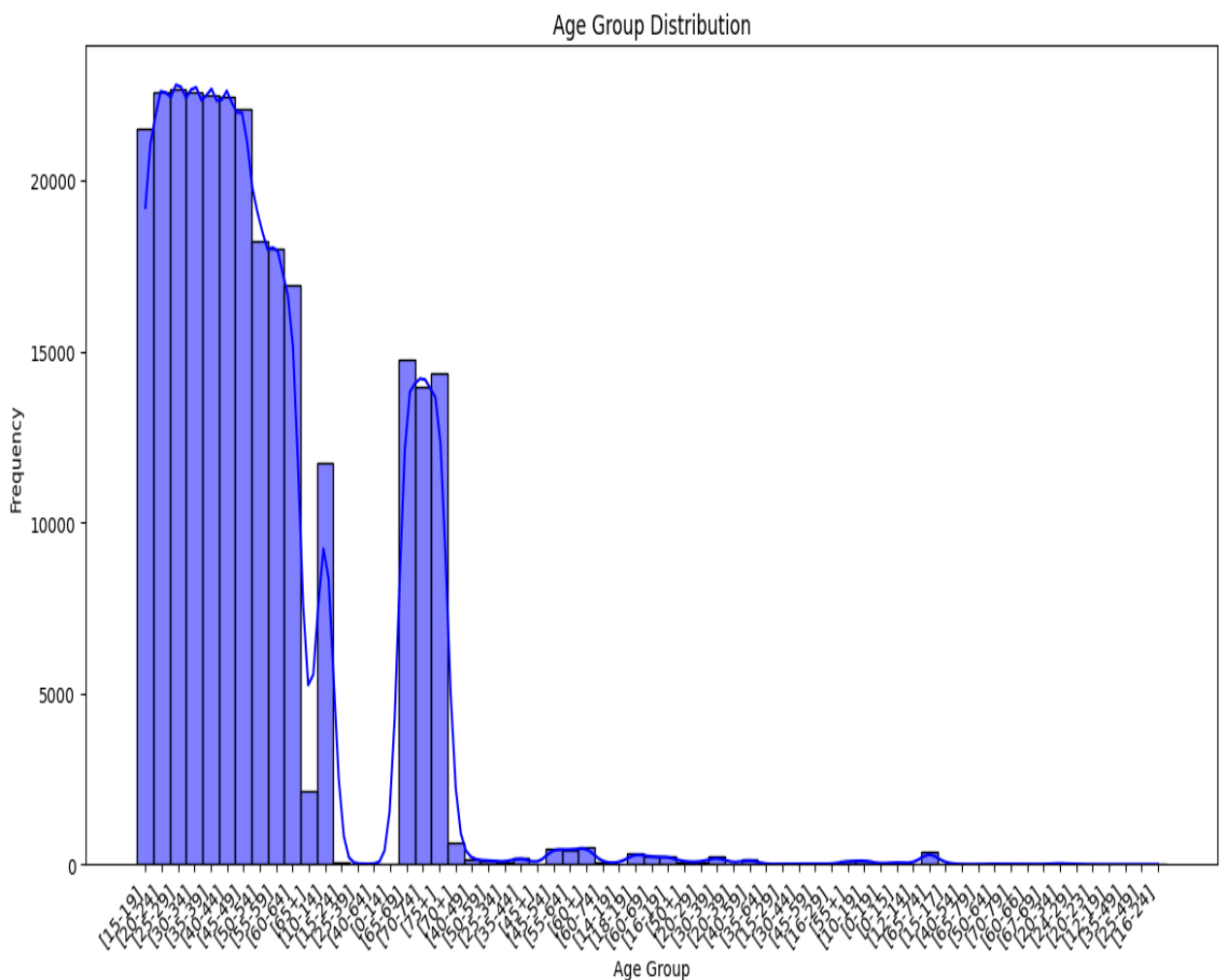
```
# Check actual column names
print("\nColumn Names:")
print(df.columns)
```

```
Column Names:
Index(['Sr.No.', 'Country', 'AgeGroup', 'Sex', 'MaritalStatus',
       'DataProcess',
       'Data Collection (Start Year)', 'Data Collection (End Year)',
       'Data Source'],
      dtype='object')
```

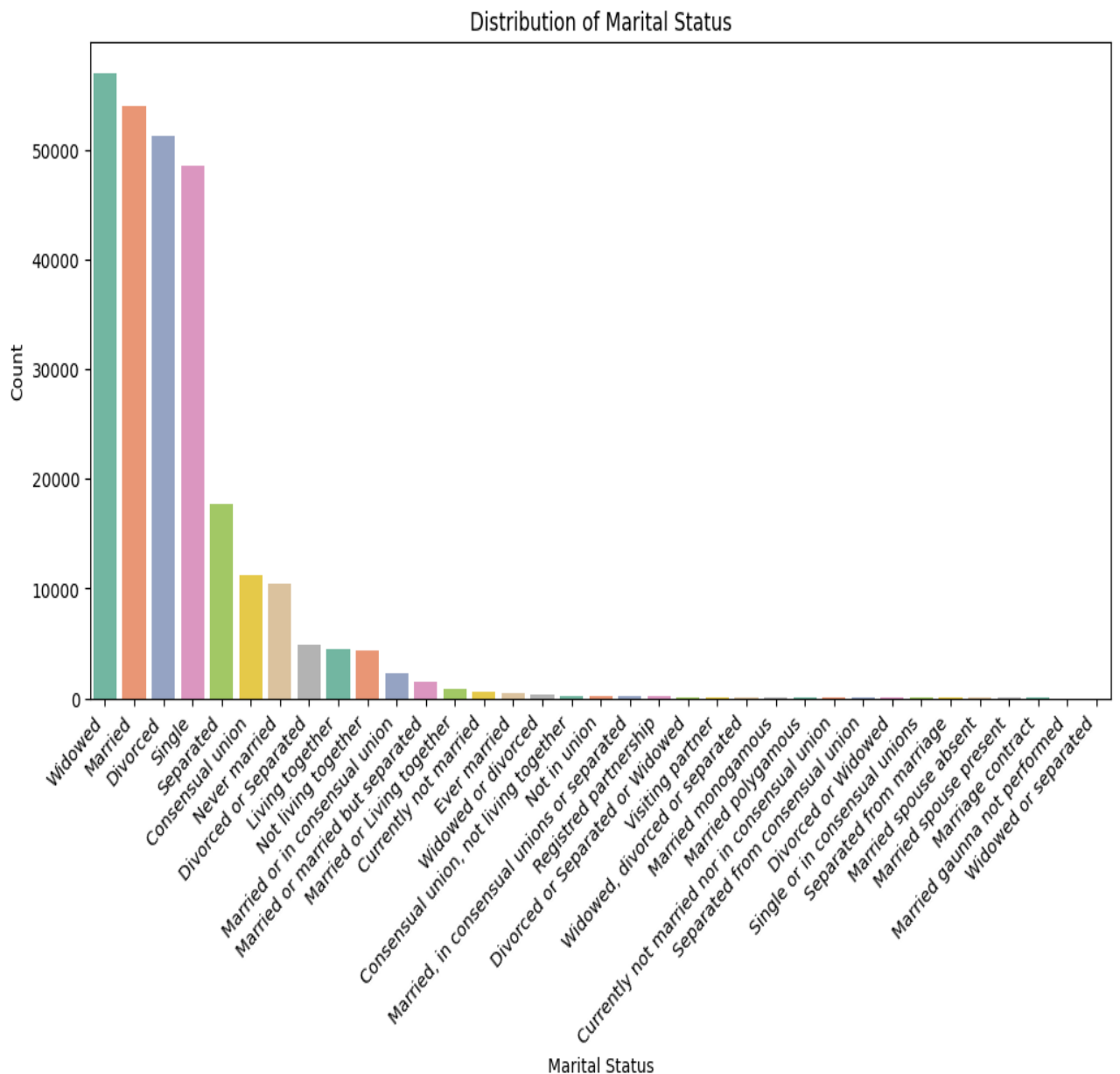
```
# Data distribution plots
if 'AgeGroup' in df.columns:
    plt.figure(figsize=(14,7))
    sns.histplot(df['AgeGroup'], bins=30, kde=True, color='blue')
    plt.title("Age Group Distribution")
```

```
plt.xlabel("Age Group")
plt.ylabel("Frequency")
plt.xticks(rotation=45, ha='right', fontsize=10)
plt.show()
else:
    print("\nColumn 'AgeGroup' not found in dataset. Please check the
column names.")
```

/usr/local/lib/python3.10/dist-packages/seaborn/_oldcore.py:1119:
FutureWarning: use_inf_as_na option is deprecated and will be removed in a
future version. Convert inf values to NaN before operating instead.
with pd.option_context('mode.use_inf_as_na', True):

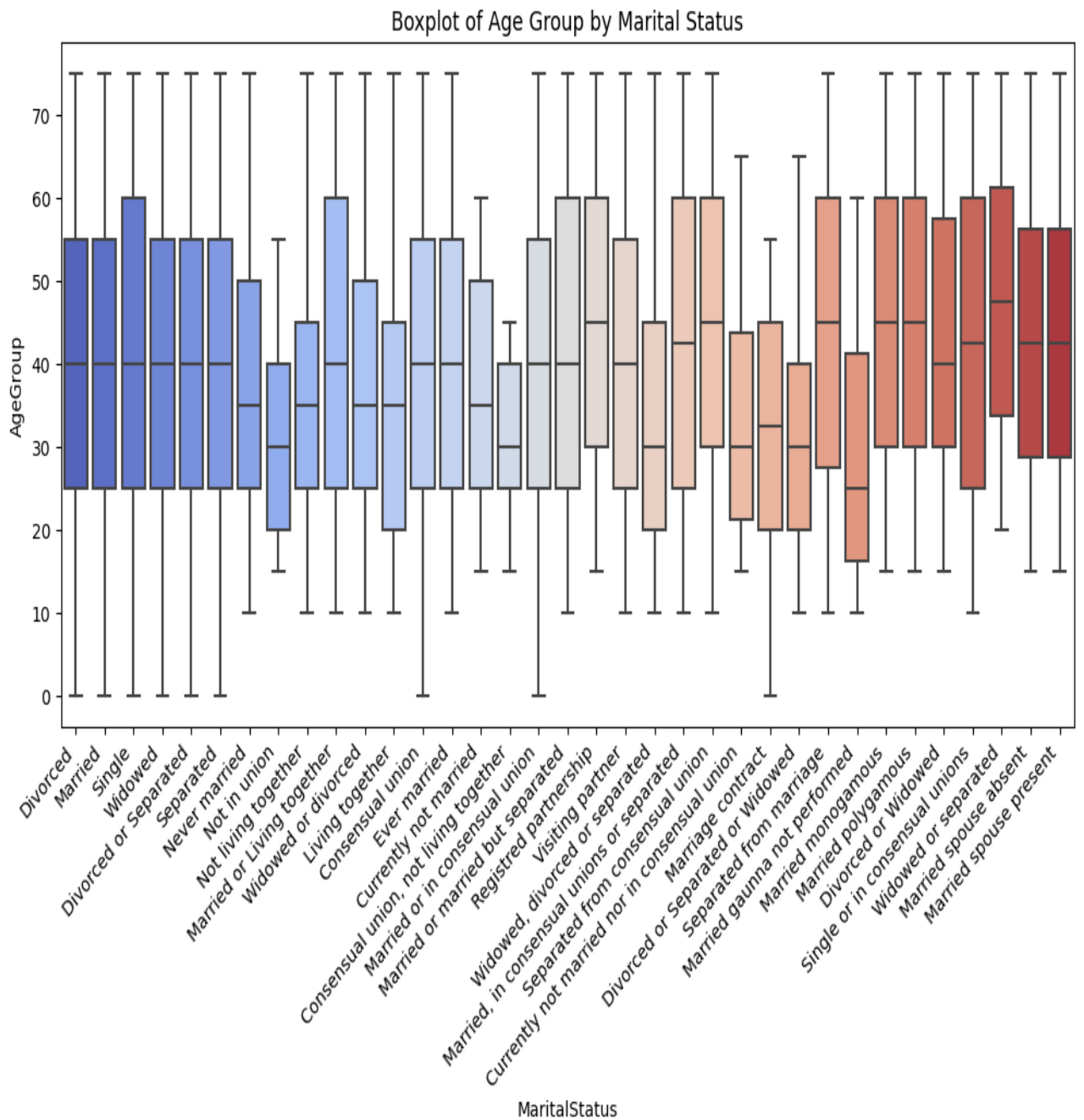


```
# Count of marital status across dataset
plt.figure(figsize=(12,6))
sns.countplot(data=df, x='MaritalStatus',
order=df['MaritalStatus'].value_counts().index, palette='Set2')
plt.title("Distribution of Marital Status")
plt.xlabel("Marital Status")
plt.ylabel("Count")
plt.xticks(rotation=45, ha='right', fontsize=10)
plt.show()
```

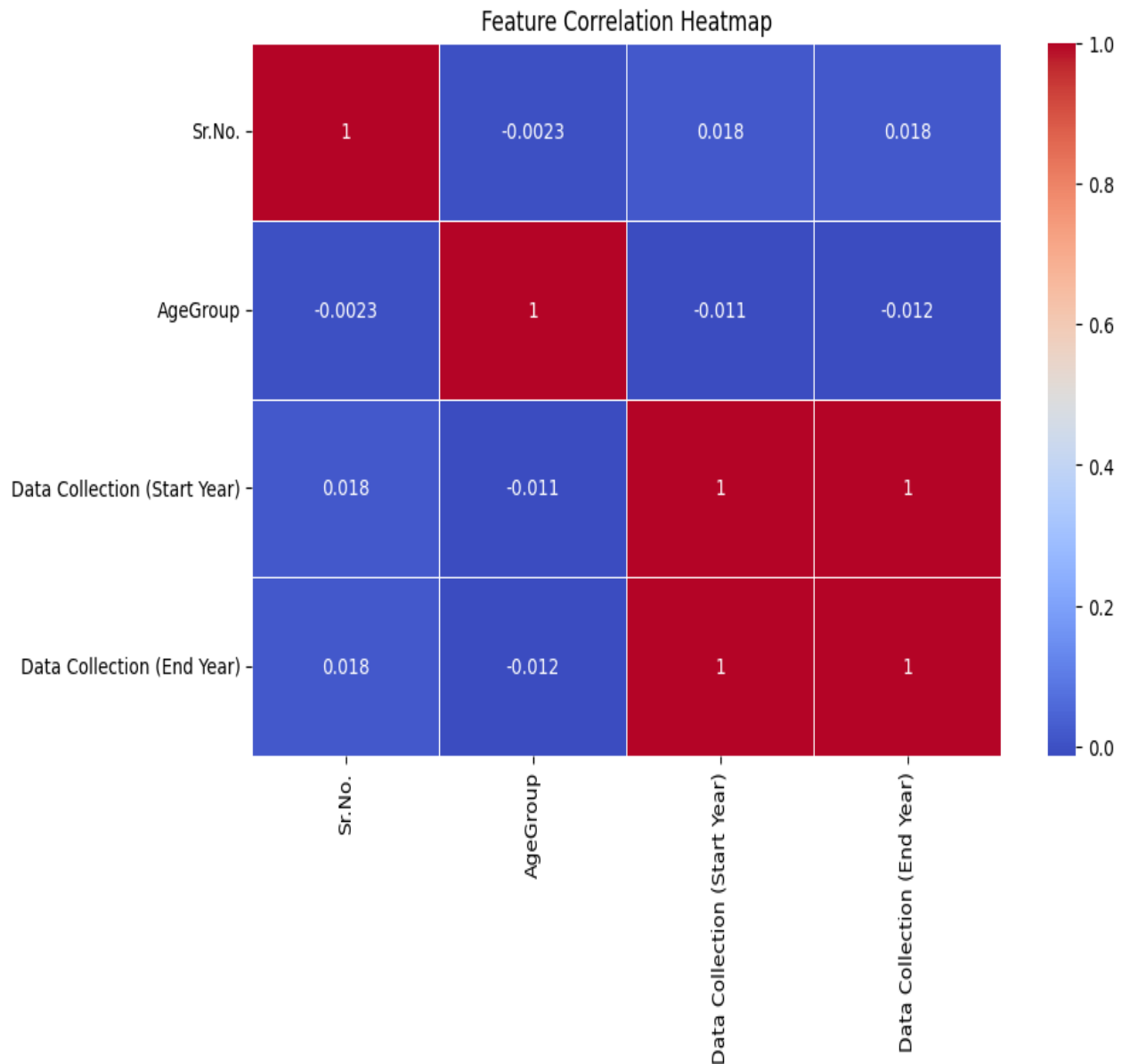


```
# Convert AgeGroup to numerical representation
df['AgeGroup'] =
df['AgeGroup'].astype(str).str.extract('(\d+)').astype(float)

# Boxplots to identify outliers in age and marital status
plt.figure(figsize=(12,6))
sns.boxplot(data=df, x='MaritalStatus', y='AgeGroup', palette='coolwarm')
plt.title("Boxplot of Age Group by Marital Status")
plt.xticks(rotation=45, ha='right', fontsize=10)
plt.show()
```



```
# Heatmap of correlations
plt.figure(figsize=(10,6))
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap='coolwarm',
linewidths=0.5)
plt.title("Feature Correlation Heatmap")
plt.show()
```



```
import seaborn as sns
import matplotlib.pyplot as plt
```

```
# Pairplot
```

```
sns.pairplot(df, hue='MaritalStatus')
plt.show()
```

```
/usr/local/lib/python3.10/dist-packages/seaborn/_oldcore.py:1119:
FutureWarning: use_inf_as_na option is deprecated and will be removed in a
future version. Convert inf values to NaN before operating instead.
```

```
    with pd.option_context('mode.use_inf_as_na', True):
```

```
/usr/local/lib/python3.10/dist-packages/seaborn/_oldcore.py:1075:
FutureWarning: When grouping with a length-1 list-like, you will need to
pass a length-1 tuple to get_group in a future version of pandas. Pass
`(name,)` instead of `name` to silence this warning.
```

```
    data_subset = grouped_data.get_group(pd_key)
```

```
/usr/local/lib/python3.10/dist-packages/seaborn/_oldcore.py:1075:
FutureWarning: When grouping with a length-1 list-like, you will need to
pass a length-1 tuple to get_group in a future version of pandas. Pass
```

```

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/usr/local/lib/python3.10/dist-packages/seaborn/_oldcore.py:1075:
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(name,) instead of name to silence this warning.
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FutureWarning: When grouping with a length-1 list-like, you will need to
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(name,) instead of name to silence this warning.
data_subset = grouped_data.get_group(pd_key)
/usr/local/lib/python3.10/dist-packages/seaborn/_oldcore.py:1075:

```

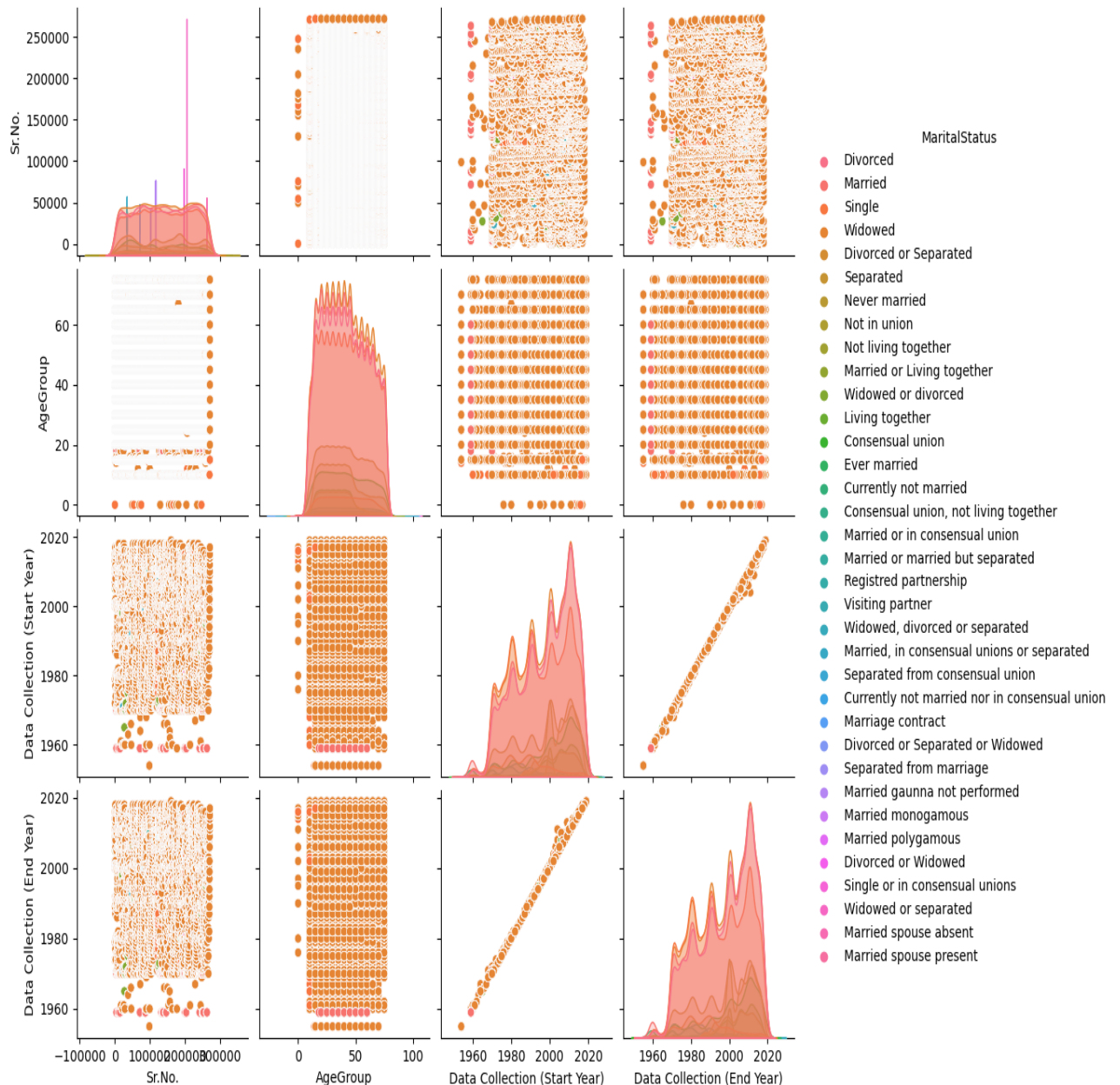
FutureWarning: When grouping with a length-1 list-like, you will need to pass a length-1 tuple to get_group in a future version of pandas. Pass `(name,)` instead of `name` to silence this warning.

```
data_subset = grouped_data.get_group(pd_key)
```

/usr/local/lib/python3.10/dist-packages/seaborn/_oldcore.py:1075:

FutureWarning: When grouping with a length-1 list-like, you will need to pass a length-1 tuple to get_group in a future version of pandas. Pass `(name,)` instead of `name` to silence this warning.

```
data_subset = grouped_data.get_group(pd_key)
```



Trend Analysis (1970-2019)

```
df['Data Collection (Start Year)'] = pd.to_numeric(df['Data Collection (Start Year)'], errors='coerce')
```

```
df['Data Collection (End Year)'] = pd.to_numeric(df['Data Collection (End Year)'], errors='coerce')
```

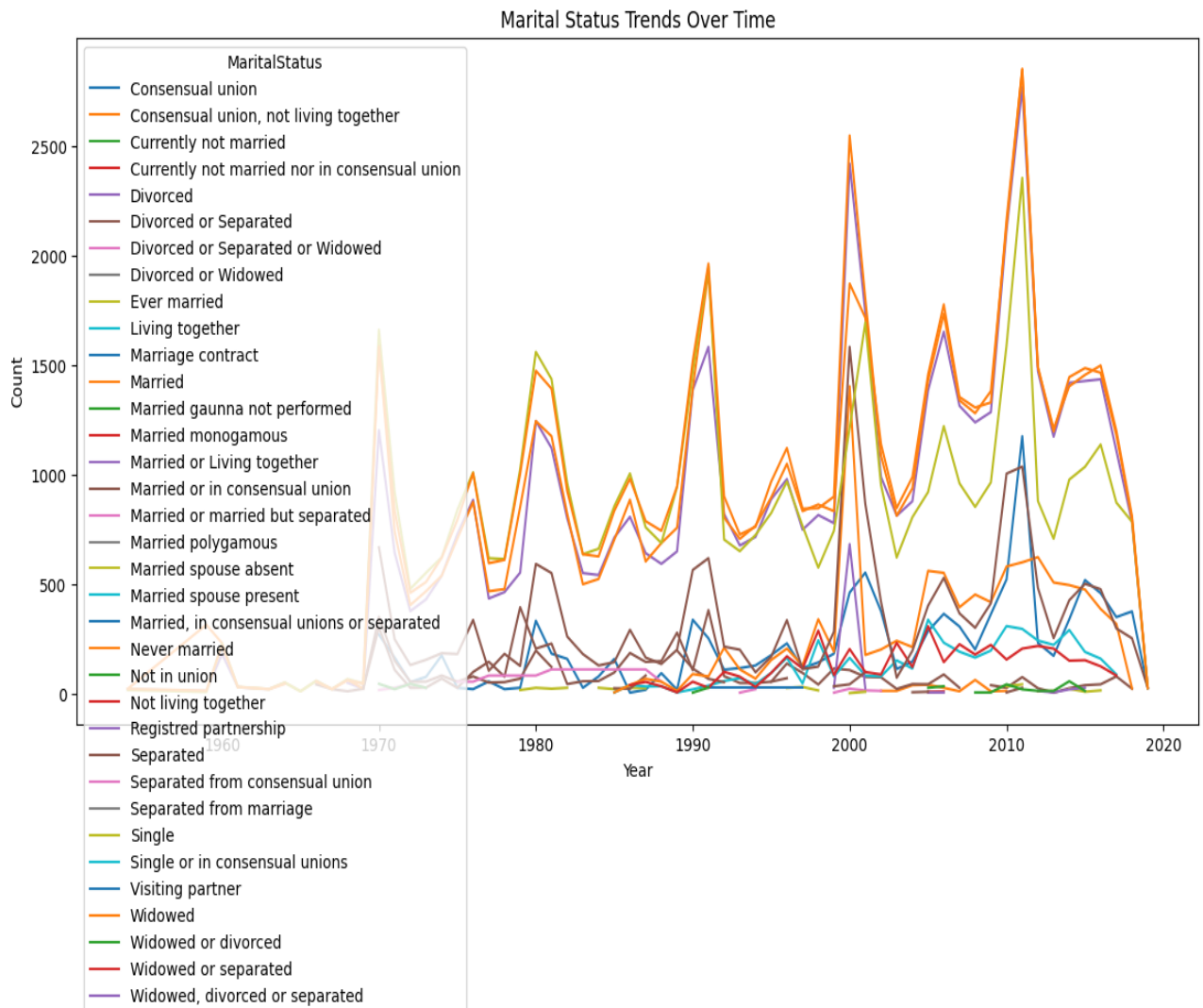
```
trend_df = df.groupby('Data Collection (Start Year)')['MaritalStatus'].value_counts().unstack()
```



```

trend_df.plot(kind='line', figsize=(14,7))
plt.title("Marital Status Trends Over Time")
plt.xlabel("Year")
plt.ylabel("Count")
plt.show()

```



```

print(df.columns)

Index(['Sr.No.', 'Country', 'AgeGroup', 'Sex', 'MaritalStatus',
      'DataProcess',
      'Data Collection (Start Year)', 'Data Collection (End Year)',
      'Data Source'],
      dtype='object')

df.rename(columns={'country': 'Country', 'marital_status':
'MaritalStatus'}, inplace=True)

import pandas as pd
df = pd.read_csv("/kaggle/input/world-marriage-dataset/World Marriage
Dataset.csv")
df['Country'] = df['Country'].astype(str)
df['MaritalStatus'] = df['MaritalStatus'].astype(str)

```



```

# Compute the number of married and divorced individuals per country
country_stats =
df.groupby('Country')['MaritalStatus'].value_counts().unstack().fillna(0)

# Sort and display the top 10 countries with the highest marriage rates
top_10_married = country_stats.sort_values(by='Married',
ascending=False).head(10)
print("Top 10 Countries with Highest Marriage Rates:")
print(top_10_married[['Married']])

# Sort and display the top 10 countries with the highest divorce rates
top_10_divorced = country_stats.sort_values(by='Divorced',
ascending=False).head(10)
print("\nTop 10 Countries with Highest Divorce Rates:")
print(top_10_divorced[['Divorced']])

# Visualization: Marriage Rates
plt.figure(figsize=(12, 6))
sns.barplot(x=top_10_married.index, y=top_10_married['Married'],
palette='Blues_r')
plt.title("Top 10 Countries with Highest Marriage Rates")
plt.xlabel("Country")
plt.ylabel("Number of Married Individuals")
plt.xticks(rotation=45)
plt.show()

# Visualization: Divorce Rates
plt.figure(figsize=(12, 6))
sns.barplot(x=top_10_divorced.index, y=top_10_divorced['Divorced'],
palette='Reds_r')
plt.title("Top 10 Countries with Highest Divorce Rates")
plt.xlabel("Country")
plt.ylabel("Number of Divorced Individuals")
plt.xticks(rotation=45)
plt.show()

```

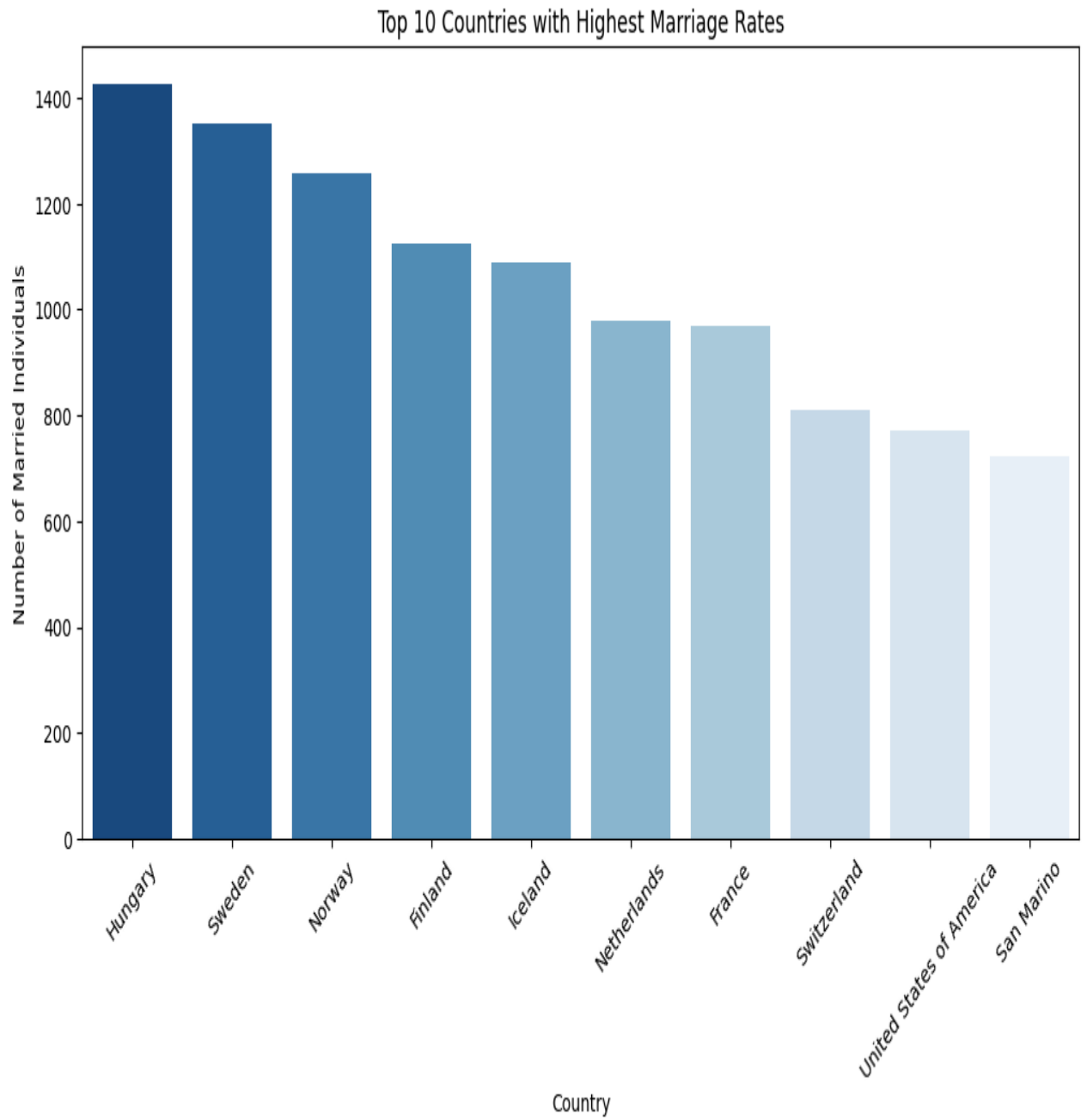
Top 10 Countries with Highest Marriage Rates:

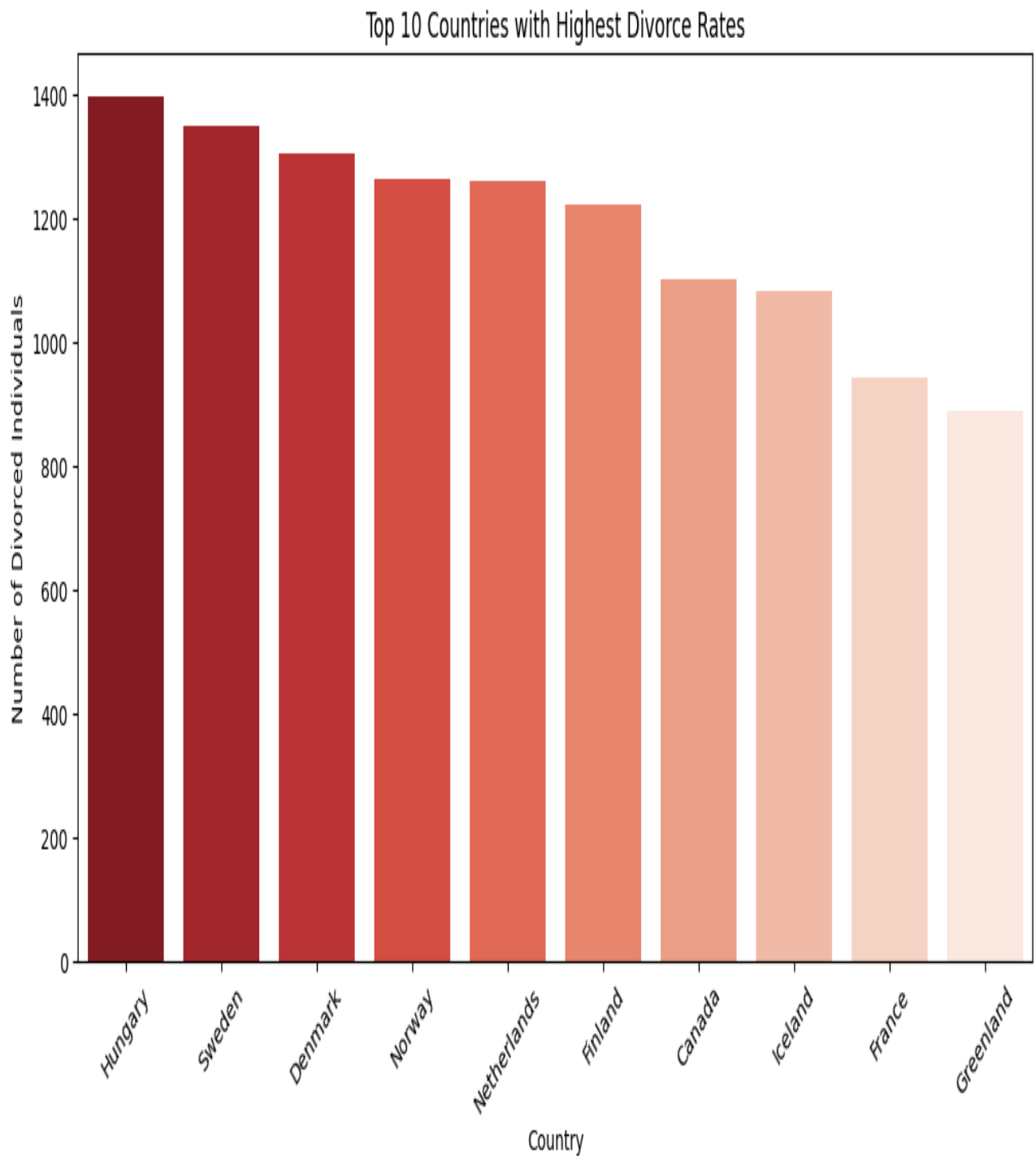
MaritalStatus	Married
Country	
Hungary	1426.0
Sweden	1351.0
Norway	1258.0
Finland	1126.0
Iceland	1088.0
Netherlands	978.0
France	970.0
Switzerland	809.0
United States of America	770.0
San Marino	723.0

Top 10 Countries with Highest Divorce Rates:

MaritalStatus	Divorced
Country	
Hungary	1398.0

Sweden	1351.0
Denmark	1305.0
Norway	1265.0
Netherlands	1260.0
Finland	1224.0
Canada	1104.0
Iceland	1084.0
France	945.0
Greenland	890.0





```
# Top 10 countries with highest and lowest marriage/divorce rates
country_stats =
df.groupby('Country')['MaritalStatus'].value_counts().unstack().fillna(0)
print("Top 10 Countries with Highest Marriage Rates:")
print(country_stats.nlargest(10, 'Married'))
print("\nTop 10 Countries with Highest Divorce Rates:")
print(country_stats.nlargest(10, 'Divorced'))
```

Top 10 Countries with Highest Marriage Rates:

MaritalStatus	Consensual union \
Country	
Hungary	344.0
Sweden	264.0
Norway	214.0
Finland	308.0
Iceland	444.0
Netherlands	180.0
France	63.0
Switzerland	194.0
United States of America	110.0
San Marino	28.0

MaritalStatus	Consensual union, not living together \
Country	
Hungary	0.0
Sweden	0.0
Norway	0.0
Finland	0.0
Iceland	0.0
Netherlands	14.0
France	26.0
Switzerland	0.0
United States of America	0.0
San Marino	0.0

MaritalStatus	Currently not married \
Country	
Hungary	0.0
Sweden	0.0
Norway	0.0
Finland	0.0
Iceland	0.0
Netherlands	0.0
France	0.0
Switzerland	0.0
United States of America	0.0
San Marino	0.0

MaritalStatus	Currently not married nor in consensual union \
Country	
Hungary	0.0
Sweden	0.0
Norway	0.0
Finland	0.0
Iceland	0.0
Netherlands	0.0
France	11.0
Switzerland	0.0
United States of America	0.0
San Marino	0.0

MaritalStatus	Divorced Divorced or Separated \
---------------	----------------------------------

Country		
Hungary	1398.0	28.0
Sweden	1351.0	0.0
Norway	1265.0	18.0
Finland	1224.0	0.0
Iceland	1084.0	0.0
Netherlands	1260.0	0.0
France	945.0	0.0
Switzerland	833.0	0.0
United States of America	782.0	0.0
San Marino	772.0	0.0

MaritalStatus	Divorced or Separated or Widowed \	
Country		
Hungary		0.0
Sweden		0.0
Norway		0.0
Finland		0.0
Iceland		0.0
Netherlands		0.0
France		12.0
Switzerland		0.0
United States of America		0.0
San Marino		0.0

MaritalStatus	Divorced or Widowed	Ever married	Living
together \			
Country			
Hungary	0.0	0.0	
0.0			
Sweden	0.0	0.0	
0.0			
Norway	0.0	0.0	
0.0			
Finland	0.0	0.0	
0.0			
Iceland	0.0	0.0	
0.0			
Netherlands	24.0	0.0	
0.0			
France	0.0	0.0	
12.0			
Switzerland	0.0	0.0	
0.0			
United States of America	0.0	0.0	
0.0			
San Marino	0.0	0.0	
0.0			

MaritalStatus	... Separated	Separated from consensual union
\		
Country	...	
Hungary	...	54.0 0.0
Sweden	...	28.0 0.0

Norway	...	1227.0	0.0
Finland	...	56.0	0.0
Iceland	...	1036.0	0.0
Netherlands	...	0.0	0.0
France	...	0.0	0.0
Switzerland	...	112.0	0.0
United States of America	...	742.0	0.0
San Marino	...	0.0	0.0

MaritalStatus	Separated from marriage	Single \
Country		
Hungary	0.0	1424.0
Sweden	0.0	1344.0
Norway	0.0	1292.0
Finland	0.0	1226.0
Iceland	0.0	1090.0
Netherlands	0.0	1284.0
France	0.0	934.0
Switzerland	0.0	838.0
United States of America	0.0	794.0
San Marino	0.0	896.0

MaritalStatus	Single or in consensual unions	Visiting partner
\		
Country		
Hungary	0.0	0.0
Sweden	0.0	0.0
Norway	0.0	0.0
Finland	0.0	0.0
Iceland	0.0	0.0
Netherlands	0.0	0.0
France	0.0	0.0
Switzerland	0.0	0.0
United States of America	0.0	0.0
San Marino	0.0	0.0

MaritalStatus	Widowed	Widowed or divorced	Widowed or
separated \			
Country			
Hungary	1424.0	0.0	
0.0			
Sweden	1348.0	0.0	
0.0			
Norway	1283.0	0.0	
0.0			
Finland	1223.0	0.0	
0.0			
Iceland	1081.0	0.0	
0.0			
Netherlands	1252.0	0.0	
0.0			
France	945.0	0.0	
0.0			
Switzerland	828.0	0.0	

0.0		
United States of America	782.0	12.0
0.0		
San Marino	884.0	0.0
0.0		

MaritalStatus	Widowed, divorced or separated
Country	
Hungary	0.0
Sweden	0.0
Norway	0.0
Finland	0.0
Iceland	0.0
Netherlands	0.0
France	0.0
Switzerland	0.0
United States of America	0.0
San Marino	0.0

[10 rows x 35 columns]

Top 10 Countries with Highest Divorce Rates:

MaritalStatus	Consensual union	Consensual union, not living together	\
Country			
Hungary	344.0		0.0
Sweden	264.0		0.0
Denmark	194.0		0.0
Norway	214.0		0.0
Netherlands	180.0		14.0
Finland	308.0		0.0
Canada	432.0		0.0
Iceland	444.0		0.0
France	63.0		26.0
Greenland	0.0		0.0

MaritalStatus	Currently not married	\
Country		
Hungary	0.0	
Sweden	0.0	
Denmark	0.0	
Norway	0.0	
Netherlands	0.0	
Finland	0.0	
Canada	0.0	
Iceland	0.0	
France	0.0	
Greenland	0.0	

MaritalStatus	Currently not married nor in consensual union	Divorced	\
Country			
Hungary	0.0	1398.0	
Sweden	0.0	1351.0	
Denmark	0.0	1305.0	
Norway	0.0	1265.0	

Netherlands	0.0	1260.0
Finland	0.0	1224.0
Canada	0.0	1104.0
Iceland	0.0	1084.0
France	11.0	945.0
Greenland	0.0	890.0

MaritalStatus	Divorced or Separated	Divorced or Separated or Widowed	\
Country			
Hungary	28.0	0.0	
Sweden	0.0	0.0	
Denmark	0.0	0.0	
Norway	18.0	0.0	
Netherlands	0.0	0.0	
Finland	0.0	0.0	
Canada	0.0	0.0	
Iceland	0.0	0.0	
France	0.0	12.0	
Greenland	28.0	0.0	

MaritalStatus	Divorced or Widowed	Ever married	Living together	...	\
Country				...	
Hungary	0.0	0.0	0.0	...	
Sweden	0.0	0.0	0.0	...	
Denmark	0.0	0.0	0.0	...	
Norway	0.0	0.0	0.0	...	
Netherlands	24.0	0.0	0.0	...	
Finland	0.0	0.0	0.0	...	
Canada	0.0	0.0	0.0	...	
Iceland	0.0	0.0	0.0	...	
France	0.0	0.0	12.0	...	
Greenland	0.0	0.0	0.0	...	

MaritalStatus	Separated	Separated from consensual union	\
Country			
Hungary	54.0	0.0	
Sweden	28.0	0.0	
Denmark	140.0	0.0	
Norway	1227.0	0.0	
Netherlands	0.0	0.0	
Finland	56.0	0.0	
Canada	332.0	0.0	
Iceland	1036.0	0.0	
France	0.0	0.0	
Greenland	56.0	0.0	

MaritalStatus	Separated from marriage	Single	\
Country			
Hungary	0.0	1424.0	
Sweden	0.0	1344.0	
Denmark	0.0	1312.0	
Norway	0.0	1292.0	
Netherlands	0.0	1284.0	
Finland	0.0	1226.0	

Canada	0.0	1154.0
Iceland	0.0	1090.0
France	0.0	934.0
Greenland	0.0	922.0

MaritalStatus	Single or in consensual unions	Visiting partner	Widowed \
Country			
Hungary	0.0	0.0	1424.0
Sweden	0.0	0.0	1348.0
Denmark	0.0	0.0	1299.0
Norway	0.0	0.0	1283.0
Netherlands	0.0	0.0	1252.0
Finland	0.0	0.0	1223.0
Canada	0.0	0.0	1104.0
Iceland	0.0	0.0	1081.0
France	0.0	0.0	945.0
Greenland	0.0	0.0	916.0

MaritalStatus	Widowed or divorced	Widowed or separated \
Country		
Hungary	0.0	0.0
Sweden	0.0	0.0
Denmark	0.0	0.0
Norway	0.0	0.0
Netherlands	0.0	0.0
Finland	0.0	0.0
Canada	36.0	0.0
Iceland	0.0	0.0
France	0.0	0.0
Greenland	0.0	0.0

MaritalStatus	Widowed, divorced or separated
Country	
Hungary	0.0
Sweden	0.0
Denmark	0.0
Norway	0.0
Netherlands	0.0
Finland	0.0
Canada	0.0
Iceland	0.0
France	0.0
Greenland	0.0

[10 rows x 35 columns]

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, classification_report
```

Load dataset

```
df = pd.read_csv("/kaggle/input/world-marriage-dataset/World Marriage
```

```

Dataset.csv") # Uncomment and replace with actual file if needed

# Drop unnecessary columns
df = df.drop(columns=['Sr.No.', 'DataProcess', 'Data Collection (Start Year)', 'Data Collection (End Year)', 'Data Source'])

# Encode categorical variables
le = LabelEncoder()
df['MaritalStatus'] = le.fit_transform(df['MaritalStatus'])
df['Sex'] = le.fit_transform(df['Sex'])
df['Country'] = le.fit_transform(df['Country'])
df['AgeGroup'] = le.fit_transform(df['AgeGroup'])

# Define features and target variable
X = df.drop(columns=['MaritalStatus']) # Features (independent variables)
y = df['MaritalStatus'] # Target variable

# Split dataset into train and test sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Train a Random Forest Classifier
model = RandomForestClassifier(n_estimators=100, random_state=42)
model.fit(X_train, y_train)

# Make predictions
y_pred = model.predict(X_test)

# Evaluate model
accuracy = accuracy_score(y_test, y_pred)
print("Model Accuracy:", accuracy)
print("Classification Report:\n", classification_report(y_test, y_pred))

```

Model Accuracy: 0.11137497468750575

Classification Report:

	precision	recall	f1-score	support
0	0.04	0.01	0.02	2260
1	0.00	0.00	0.00	60
2	0.08	0.02	0.03	101
3	0.00	0.00	0.00	13
4	0.10	0.10	0.10	10107
5	0.03	0.00	0.01	987
6	0.00	0.00	0.00	26
7	0.00	0.00	0.00	5
8	0.17	0.01	0.02	103
9	0.00	0.00	0.00	926
10	0.00	0.00	0.00	4
11	0.11	0.13	0.12	10909
12	0.00	0.00	0.00	2
13	0.00	0.00	0.00	10
14	0.00	0.00	0.00	174
15	0.09	0.01	0.01	454
16	0.00	0.00	0.00	323

17	0.00	0.00	0.00	10
18	0.00	0.00	0.00	7
19	0.00	0.00	0.00	5
20	0.00	0.00	0.00	40
21	0.02	0.00	0.00	2046
22	0.00	0.00	0.00	45
23	0.00	0.00	0.00	802
24	0.00	0.00	0.00	41
25	0.05	0.02	0.03	3581
26	0.00	0.00	0.00	3
27	0.00	0.00	0.00	14
28	0.14	0.16	0.15	9646
29	0.00	0.00	0.00	7
30	0.00	0.00	0.00	17
31	0.12	0.17	0.14	11516
32	0.00	0.00	0.00	61
33	0.00	0.00	0.00	6
34	0.00	0.00	0.00	10
accuracy				0.11 54321
macro avg		0.03 0.02 0.02	54321	
weighted avg		0.10 0.11 0.10	54321	

```
/usr/local/lib/python3.10/dist-
packages/sklearn/metrics/_classification.py:1344: UndefinedMetricWarning:
Precision and F-score are ill-defined and being set to 0.0 in labels with
no predicted samples. Use `zero_division` parameter to control this
behavior.
```

```
_warn_prf(average, modifier, msg_start, len(result))
```

```
/usr/local/lib/python3.10/dist-
packages/sklearn/metrics/_classification.py:1344: UndefinedMetricWarning:
Precision and F-score are ill-defined and being set to 0.0 in labels with
no predicted samples. Use `zero_division` parameter to control this
behavior.
```

```
_warn_prf(average, modifier, msg_start, len(result))
```

```
/usr/local/lib/python3.10/dist-
packages/sklearn/metrics/_classification.py:1344: UndefinedMetricWarning:
Precision and F-score are ill-defined and being set to 0.0 in labels with
no predicted samples. Use `zero_division` parameter to control this
behavior.
```

```
_warn_prf(average, modifier, msg_start, len(result))
```

```
# Logistic Regression
```

```
logreg = LogisticRegression()
```

```
logreg.fit(X_train, y_train)
```

```
y_pred = logreg.predict(X_test)
```

```
print("\nLogistic Regression Accuracy:", accuracy_score(y_test, y_pred))
```

```
print(classification_report(y_test, y_pred))
```

```
Logistic Regression Accuracy: 0.2119990427274903
```

```
precision recall f1-score support
```

```
0 0.00 0.00 0.00 2260
```

1	0.00	0.00	0.00	60
2	0.00	0.00	0.00	101
3	0.00	0.00	0.00	13
4	0.00	0.00	0.00	10107
5	0.00	0.00	0.00	987
6	0.00	0.00	0.00	26
7	0.00	0.00	0.00	5
8	0.00	0.00	0.00	103
9	0.00	0.00	0.00	926
10	0.00	0.00	0.00	4
11	0.00	0.00	0.00	10909
12	0.00	0.00	0.00	2
13	0.00	0.00	0.00	10
14	0.00	0.00	0.00	174
15	0.00	0.00	0.00	454
16	0.00	0.00	0.00	323
17	0.00	0.00	0.00	10
18	0.00	0.00	0.00	7
19	0.00	0.00	0.00	5
20	0.00	0.00	0.00	40
21	0.00	0.00	0.00	2046
22	0.00	0.00	0.00	45
23	0.00	0.00	0.00	802
24	0.00	0.00	0.00	41
25	0.00	0.00	0.00	3581
26	0.00	0.00	0.00	3
27	0.00	0.00	0.00	14
28	0.00	0.00	0.00	9646
29	0.00	0.00	0.00	7
30	0.00	0.00	0.00	17
31	0.21	1.00	0.35	11516
32	0.00	0.00	0.00	61
33	0.00	0.00	0.00	6
34	0.00	0.00	0.00	10
accuracy				0.21 54321
macro avg				0.01 0.03 0.01 54321
weighted avg				0.04 0.21 0.07 54321

```

/usr/local/lib/python3.10/dist-
packages/sklearn/linear_model/_logistic.py:458: ConvergenceWarning: lbfgs
failed to converge (status=1):
STOP: TOTAL NO. of ITERATIONS REACHED LIMIT.

```

Increase the number of iterations (max_iter) or scale the data as shown in:

<https://scikit-learn.org/stable/modules/preprocessing.html>

Please also refer to the documentation for alternative solver options:

https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression

```
n_iter_i = _check_optimize_result(
```

```
/usr/local/lib/python3.10/dist-
```

```

packages/sklearn/metrics/_classification.py:1344: UndefinedMetricWarning:
Precision and F-score are ill-defined and being set to 0.0 in labels with
no predicted samples. Use `zero division` parameter to control this

```

behavior.

```
_warn_prf(average, modifier, msg_start, len(result))
/usr/local/lib/python3.10/dist-
packages/sklearn/metrics/_classification.py:1344: UndefinedMetricWarning:
Precision and F-score are ill-defined and being set to 0.0 in labels with
no predicted samples. Use `zero_division` parameter to control this
behavior.
```

```
_warn_prf(average, modifier, msg_start, len(result))
/usr/local/lib/python3.10/dist-
packages/sklearn/metrics/_classification.py:1344: UndefinedMetricWarning:
Precision and F-score are ill-defined and being set to 0.0 in labels with
no predicted samples. Use `zero_division` parameter to control this
behavior.
```

```
_warn_prf(average, modifier, msg_start, len(result))
```

Decision Tree

```
dt = DecisionTreeClassifier()
dt.fit(X_train, y_train)
y_pred_dt = dt.predict(X_test)
print("\nDecision Tree Accuracy:", accuracy_score(y_test, y_pred_dt))
print(classification_report(y_test, y_pred_dt))
```

Decision Tree Accuracy: 0.1076195209955634

	precision	recall	f1-score	support
0	0.03	0.02	0.02	2260
1	0.00	0.00	0.00	60
2	0.15	0.06	0.09	101
3	0.00	0.00	0.00	13
4	0.10	0.14	0.12	10107
5	0.04	0.01	0.01	987
6	0.00	0.00	0.00	26
7	0.00	0.00	0.00	5
8	0.20	0.03	0.05	103
9	0.03	0.00	0.00	926
10	0.00	0.00	0.00	4
11	0.11	0.14	0.12	10909
12	0.00	0.00	0.00	2
13	0.00	0.00	0.00	10
14	0.00	0.00	0.00	174
15	0.06	0.00	0.01	454
16	0.00	0.00	0.00	323
17	0.00	0.00	0.00	10
18	0.00	0.00	0.00	7
19	0.00	0.00	0.00	5
20	0.00	0.00	0.00	40
21	0.02	0.00	0.00	2046
22	0.00	0.00	0.00	45
23	0.00	0.00	0.00	802
24	0.00	0.00	0.00	41
25	0.05	0.02	0.03	3581
26	0.00	0.00	0.00	3
27	0.00	0.00	0.00	14
28	0.14	0.14	0.14	9646

29	0.00	0.00	0.00	7
30	0.00	0.00	0.00	17
31	0.11	0.11	0.11	11516
32	0.00	0.00	0.00	61
33	0.00	0.00	0.00	6
34	0.00	0.00	0.00	10
accuracy			0.11	54321
macro avg	0.03	0.02	0.02	54321
weighted avg	0.10	0.11	0.10	54321

```
/usr/local/lib/python3.10/dist-
packages/sklearn/metrics/_classification.py:1344: UndefinedMetricWarning:
Precision and F-score are ill-defined and being set to 0.0 in labels with
no predicted samples. Use `zero_division` parameter to control this
behavior.
```

```
_warn_prf(average, modifier, msg_start, len(result))
```

```
/usr/local/lib/python3.10/dist-
packages/sklearn/metrics/_classification.py:1344: UndefinedMetricWarning:
Precision and F-score are ill-defined and being set to 0.0 in labels with
no predicted samples. Use `zero_division` parameter to control this
behavior.
```

```
_warn_prf(average, modifier, msg_start, len(result))
```

```
/usr/local/lib/python3.10/dist-
packages/sklearn/metrics/_classification.py:1344: UndefinedMetricWarning:
Precision and F-score are ill-defined and being set to 0.0 in labels with
no predicted samples. Use `zero_division` parameter to control this
behavior.
```

```
_warn_prf(average, modifier, msg_start, len(result))
```

```
# Random Forest
```

```
rf = RandomForestClassifier()
```

```
rf.fit(X_train, y_train)
```

```
y_pred_rf = rf.predict(X_test)
```

```
print("\nRandom Forest Accuracy:", accuracy_score(y_test, y_pred_rf))
```

```
print(classification_report(y_test, y_pred_rf))
```

```
Random Forest Accuracy: 0.11166952007510908
```

	precision	recall	f1-score	support
0	0.03	0.01	0.02	2260
1	0.00	0.00	0.00	60
2	0.10	0.02	0.03	101
3	0.00	0.00	0.00	13
4	0.10	0.10	0.10	10107
5	0.01	0.00	0.00	987
6	0.00	0.00	0.00	26
7	0.00	0.00	0.00	5
8	0.17	0.01	0.02	103
9	0.00	0.00	0.00	926
10	0.00	0.00	0.00	4
11	0.11	0.13	0.12	10909
12	0.00	0.00	0.00	2

13	0.00	0.00	0.00	10
14	0.00	0.00	0.00	174
15	0.05	0.00	0.01	454
16	0.00	0.00	0.00	323
17	0.00	0.00	0.00	10
18	0.00	0.00	0.00	7
19	0.00	0.00	0.00	5
20	0.00	0.00	0.00	40
21	0.01	0.00	0.00	2046
22	0.00	0.00	0.00	45
23	0.00	0.00	0.00	802
24	0.00	0.00	0.00	41
25	0.05	0.02	0.03	3581
26	0.00	0.00	0.00	3
27	0.00	0.00	0.00	14
28	0.14	0.16	0.15	9646
29	0.00	0.00	0.00	7
30	0.00	0.00	0.00	17
31	0.12	0.17	0.14	11516
32	0.00	0.00	0.00	61
33	0.00	0.00	0.00	6
34	0.00	0.00	0.00	10
accuracy				0.11 54321
macro avg				0.03 0.02 0.02 54321
weighted avg				0.10 0.11 0.10 54321

```
/usr/local/lib/python3.10/dist-
packages/sklearn/metrics/_classification.py:1344: UndefinedMetricWarning:
Precision and F-score are ill-defined and being set to 0.0 in labels with
no predicted samples. Use `zero_division` parameter to control this
behavior.
```

```
_warn_prf(average, modifier, msg_start, len(result))
/usr/local/lib/python3.10/dist-
packages/sklearn/metrics/_classification.py:1344: UndefinedMetricWarning:
Precision and F-score are ill-defined and being set to 0.0 in labels with
no predicted samples. Use `zero_division` parameter to control this
behavior.
```

```
_warn_prf(average, modifier, msg_start, len(result))
/usr/local/lib/python3.10/dist-
packages/sklearn/metrics/_classification.py:1344: UndefinedMetricWarning:
Precision and F-score are ill-defined and being set to 0.0 in labels with
no predicted samples. Use `zero_division` parameter to control this
behavior.
```

```
_warn_prf(average, modifier, msg_start, len(result))
```

```
# Clustering (K-Means)
```

```
kmeans = KMeans(n_clusters=3, random_state=42)
```

```
kmeans.fit(X)
```

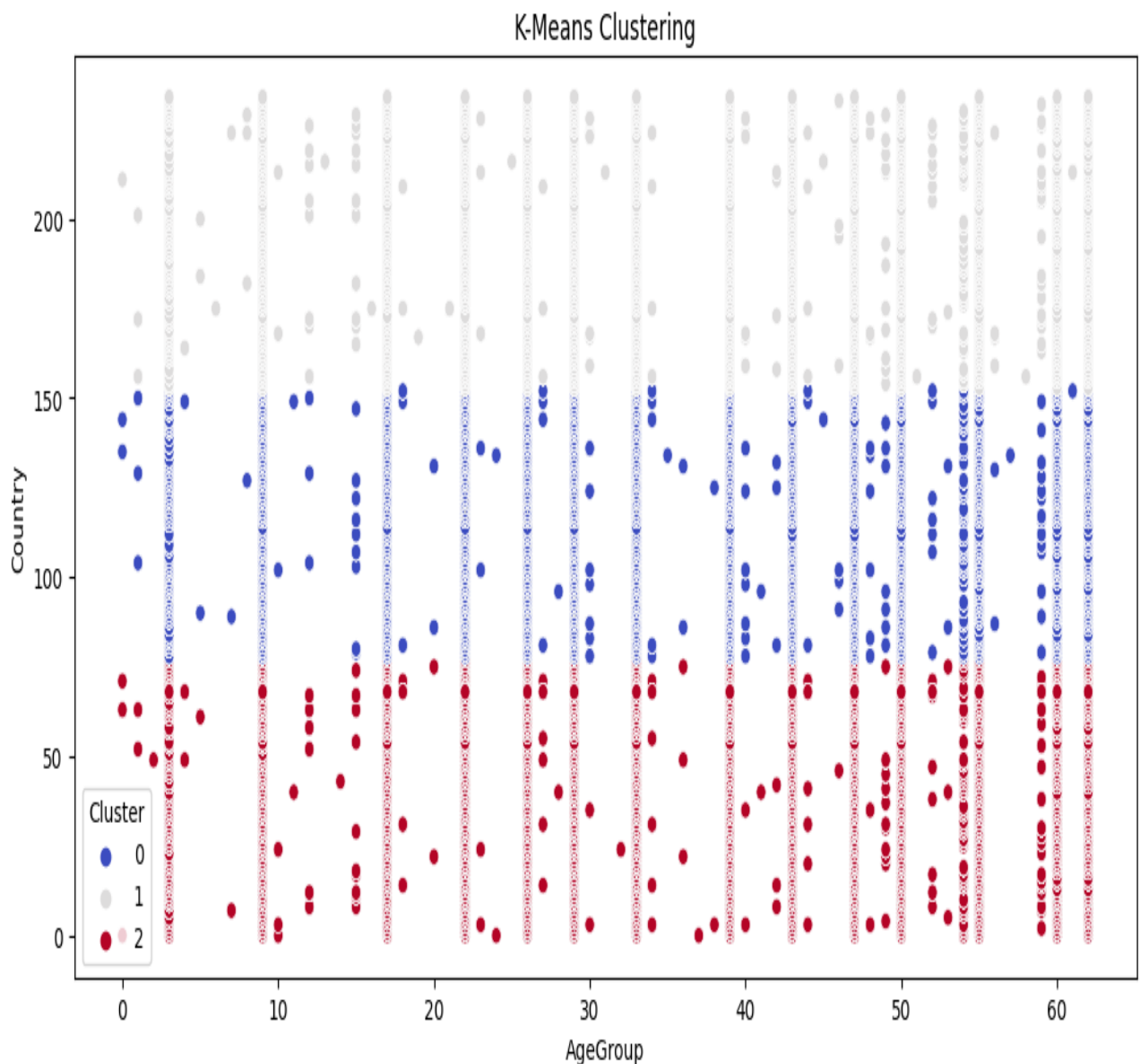
```
df['Cluster'] = kmeans.labels_
```

```
plt.figure(figsize=(12,6))
```

```
sns.scatterplot(data=df, x='AgeGroup', y='Country', hue='Cluster',
palette='coolwarm')
```

```
plt.title("K-Means Clustering")
plt.show()
```

```
/usr/local/lib/python3.10/dist-packages/sklearn/cluster/_kmeans.py:870:
FutureWarning: The default value of `n_init` will change from 10 to 'auto'
in 1.4. Set the value of `n_init` explicitly to suppress the warning
    warnings.warn(
/usr/local/lib/python3.10/dist-packages/IPython/core/pylabtools.py:151:
UserWarning: Creating legend with loc="best" can be slow with large amounts
of data.
    fig.canvas.print_figure(bytes_io, **kw)
```



```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.cluster import KMeans, DBSCAN
```



```

# Load dataset (Ensure the file path is correct)
df = pd.read_csv(r"/kaggle/input/world-marriage-dataset/World Marriage
Dataset.csv")

# Drop unnecessary columns
df = df.drop(columns=['Sr.No.', 'DataProcess', 'Data Collection (Start
Year)', 'Data Collection (End Year)', 'Data Source'])

# Encode categorical variables
le = LabelEncoder()
df['MaritalStatus_encoded'] = le.fit_transform(df['MaritalStatus'])
df['Sex'] = le.fit_transform(df['Sex'])
df['Country'] = le.fit_transform(df['Country'])
df['AgeGroup'] = le.fit_transform(df['AgeGroup'])

# Define features for clustering
X = df[['AgeGroup', 'Sex', 'Country', 'MaritalStatus_encoded']]

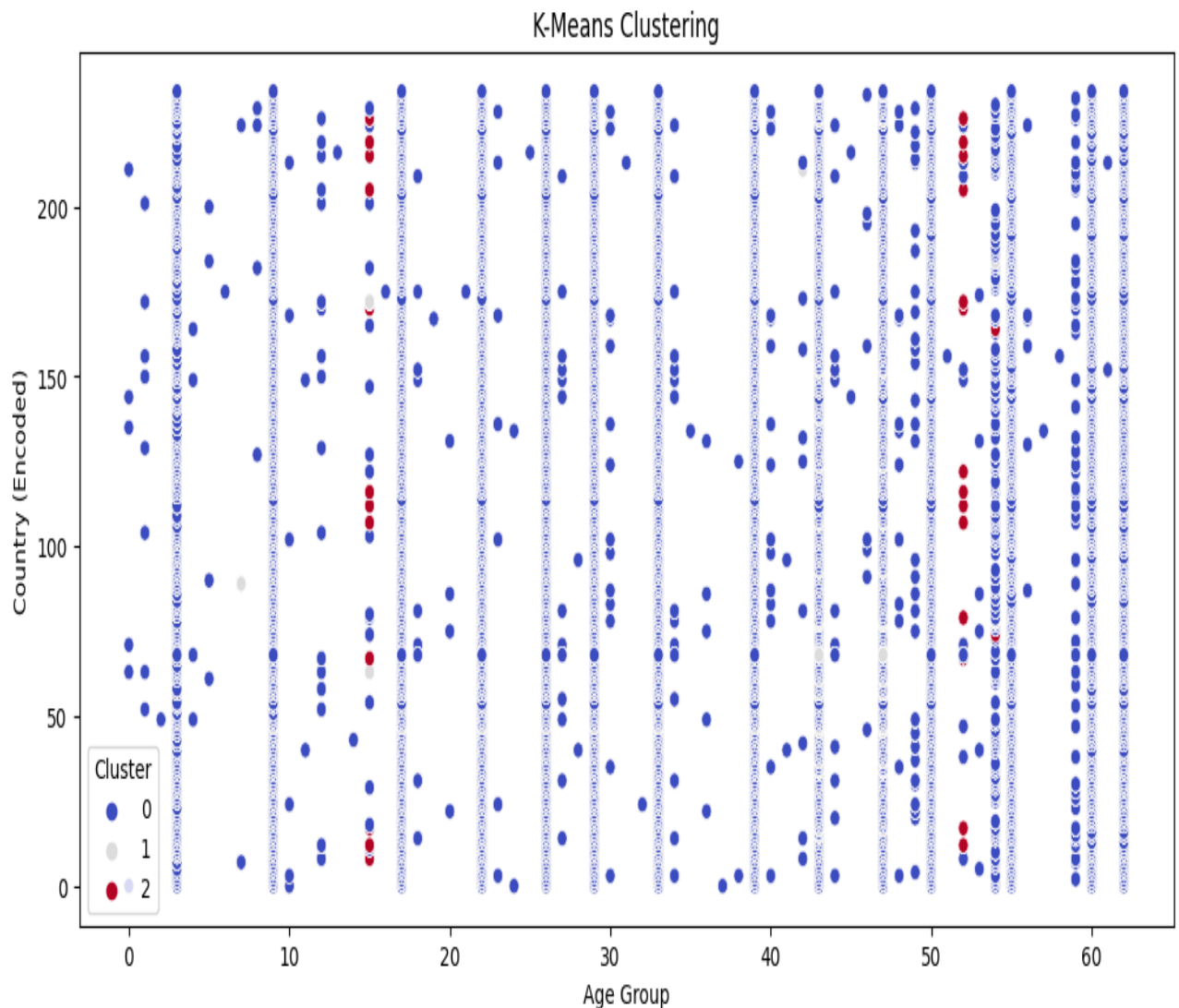
# Scale features (important for clustering)
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

### K-Means Clustering
kmeans = KMeans(n_clusters=3, random_state=42, n_init=10)
kmeans.fit(X_scaled)
df['KMeans_Cluster'] = kmeans.labels_

# Visualizing K-Means Clusters
plt.figure(figsize=(12, 6))
sns.scatterplot(data=df, x='AgeGroup', y='Country', hue='KMeans_Cluster',
palette='coolwarm')
plt.title("K-Means Clustering")
plt.xlabel("Age Group")
plt.ylabel("Country (Encoded)")
plt.legend(title="Cluster")
plt.show()

/usr/local/lib/python3.10/dist-packages/IPython/core/pylabtools.py:151:
UserWarning: Creating legend with loc="best" can be slow with large amounts
of data.
  fig.canvas.print_figure(bytes_io, **kw)

```



```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.cluster import DBSCAN

# Load dataset
df = pd.read_csv("/kaggle/input/world-marriage-dataset/World Marriage
Dataset.csv") # Uncomment and replace with actual file if needed

# Drop unnecessary columns
df = df.drop(columns=['Sr.No.', 'DataProcess', 'Data Collection (Start
Year)', 'Data Collection (End Year)', 'Data Source'])

# Encode categorical variables
le = LabelEncoder()
df['MaritalStatus_encoded'] = le.fit_transform(df['MaritalStatus'])
df['Sex'] = le.fit_transform(df['Sex'])
df['Country'] = le.fit_transform(df['Country'])
df['AgeGroup'] = le.fit_transform(df['AgeGroup'])
```

```

# Define features for clustering
X = df[['AgeGroup', 'Sex', 'Country', 'MaritalStatus_encoded']]

# Scale features (important for DBSCAN)
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

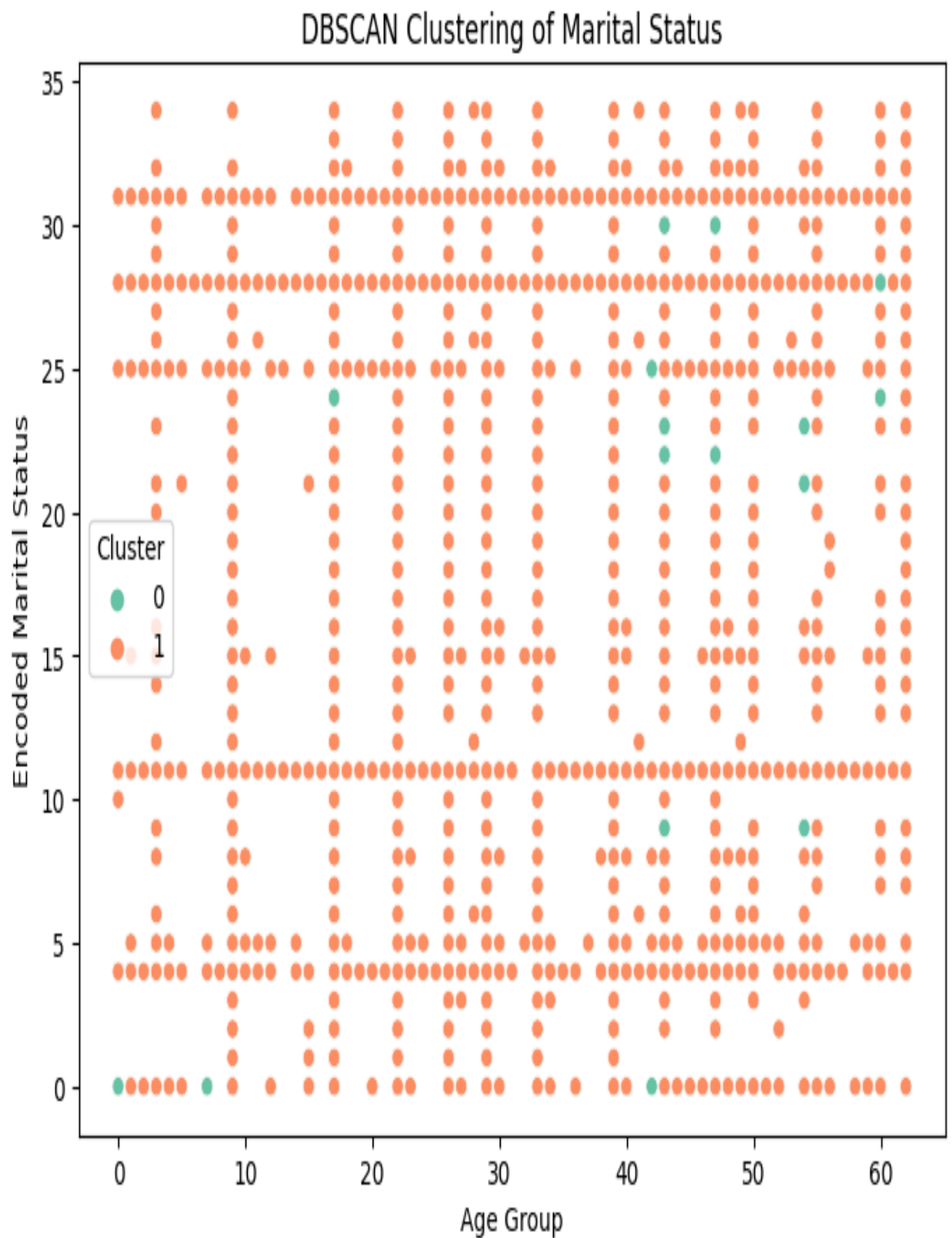
# Apply DBSCAN Clustering
dbscan = DBSCAN(eps=0.5, min_samples=5)
df['DBSCAN_Cluster'] = dbscan.fit_predict(X_scaled)

# Visualizing the clusters
plt.figure(figsize=(8, 6))
sns.scatterplot(x=df['AgeGroup'], y=df['MaritalStatus_encoded'],
hue=df['DBSCAN_Cluster'], palette='Set2')
plt.title("DBSCAN Clustering of Marital Status")
plt.xlabel("Age Group")
plt.ylabel("Encoded Marital Status")
plt.legend(title="Cluster")
plt.show()

print("\n✔Project Analysis Completed - By Nadeem Ahamad")

/usr/local/lib/python3.10/dist-packages/IPython/core/pylabtools.py:151:
UserWarning: Creating legend with loc="best" can be slow with large amounts
of data.
  fig.canvas.print_figure(bytes_io, **kw)

```



✔ Project Analysis Completed - By Nadeem Ahamad

Dashboard Code

```
# ----- Enhanced Streamlit Marriage Dataset Dashboard with Background
-----

# Import Libraries
import streamlit as st
import pandas as pd
import plotly.express as px
from sklearn.preprocessing import LabelEncoder
from sklearn.cluster import KMeans

# ----- Streamlit Page Config -----
st.set_page_config(page_title="World Marriage Insights", layout="wide")

# ----- Custom Background & Styling -----
st.markdown(
    """
    <style>
    /* Background Image with Gradient Overlay */
    .stApp {
        background: linear-gradient(135deg, rgba(217, 167, 199, 0.7),
        rgba(255, 252, 220, 0.7)),
        url("https://source.unsplash.com/1600x900/?traditional,marriage,culture");
        background-size: cover;
        background-attachment: fixed;
        background-position: center;
    }

    /* Frosted Glass Effect for Components */
    .css-18e3th9 {
        background: rgba(255, 255, 255, 0.8); /* semi-transparent white */
        backdrop-filter: blur(8px);
        border-radius: 20px;
        padding: 20px;
    }

    /* Sidebar Styling */
    .css-1d391kg {
        background: rgba(255, 255, 255, 0.85); /* Adjust opacity */
        backdrop-filter: blur(6px);
        border-radius: 15px;
        padding: 16px;
    }

    /* Text Color Adjustments */
    h1, h2, h3, h4, h5, h6, p, label {
        color: #333333; /* Dark gray */
    }
    </style>
    """,
    unsafe_allow_html=True
)
```

```

# ----- Load Dataset -----
@st.cache_data
def load_data():
    # FIXED path with raw string
    df = pd.read_csv("D:\MCA Life 2023-25\College D Y Patil\Sem 4
Project\Major Project Final Submission\World_Marriage_Dashboard_Fixed\World
Marriage Dataset till 2017.csv")

    return df

df = load_data()

# ----- Title & User Guidance -----
st.title("🌐 World Marriage Dataset - Global Trends & Insights (1970-2017)")

st.markdown("""
Welcome to the Interactive Dashboard on Global Marriage Trends.
Here you can explore, analyze, and discover patterns in marriage,
divorce, and other marital statuses across 232 countries over 5 decades.
📌 Tip: Use the sidebar to filter data dynamically by Country, Age
Group, Gender, Marital Status to get personalized insights.
""")

# ----- Sidebar Filters -----
st.sidebar.header("🌐 Explore Data Filters")
countries = st.sidebar.multiselect("Select Countries",
options=df['Country'].unique(),
                                default=df['Country'].unique()[:10])
age_groups = st.sidebar.multiselect("Select Age Groups",
options=df['AgeGroup'].unique(),
                                default=df['AgeGroup'].unique())
genders = st.sidebar.multiselect("Select Gender",
options=df['Sex'].unique(), default=df['Sex'].unique())
marital_statuses = st.sidebar.multiselect("Select Marital Status",
options=df['MaritalStatus'].unique(),
                                default=df['MaritalStatus'].unique())

# ----- Filter Dataset -----
filtered_df = df[
    (df['Country'].isin(countries)) &
    (df['AgeGroup'].isin(age_groups)) &
    (df['Sex'].isin(genders)) &
    (df['MaritalStatus'].isin(marital_statuses))
]

# ----- Dynamic Insights (Key Stats) -----
st.header("📊 Key Insights from Filtered Data")

if not filtered_df.empty:
    total_records = len(filtered_df)

```

```

    married_count = len(filtered_df[filtered_df['MaritalStatus'] ==
'Married'])
    divorced_count = len(filtered_df[filtered_df['MaritalStatus'] ==
'Divorced'])
    most_common_status = filtered_df['MaritalStatus'].mode()[0]
    top_country_married = filtered_df[filtered_df['MaritalStatus'] ==
'Married']['Country'].mode()[0]

    # Stat cards
    col1, col2, col3, col4 = st.columns(4)
    col1.metric("Total Records", total_records)
    col2.metric("Total Married", married_count)
    col3.metric("Total Divorced", divorced_count)
    col4.metric("Most Common Status", most_common_status)

    st.markdown(f"📊 Top Country for Marriage (in filtered data):
**{top_country_married}**")
    else:
        st.warning("📊 No data available for the selected filters. Please adjust
filters to see insights.")

    # ----- Filtered Data Preview -----
    with st.expander("📊 Click to View Filtered Data (First 50 Rows)"):
        st.dataframe(filtered_df.head(50))

    # ----- Visualization Section -----

    # 1. Marital Status Distribution
    st.subheader("📊 Marital Status Distribution by Gender")
    fig1 = px.histogram(filtered_df, x='MaritalStatus', color='Sex',
barmode='group',
                        title="Distribution of Marital Status", text_auto=True)
    st.plotly_chart(fig1, use_container_width=True)

    # 2. Top Marriage & Divorce Rates
    st.subheader("📊 Country-wise Marriage & Divorce Rates")
    country_stats =
filtered_df.groupby('Country')['MaritalStatus'].value_counts().unstack().fi
llna(0)

    if 'Married' in country_stats.columns:
        top_married = country_stats['Married'].nlargest(10)
        st.markdown(f"✅ **{top_married.idxmax()}** has the highest number of
married individuals in the selected data.")
        fig2 = px.bar(top_married, x=top_married.index, y=top_married.values,
                        title='Top 10 Countries with Highest Marriage Rates',
                        labels={'y': 'Number of Married Individuals', 'x':
'Country'}, text_auto=True)
        st.plotly_chart(fig2, use_container_width=True)

    if 'Divorced' in country_stats.columns:
        top_divorced = country_stats['Divorced'].nlargest(10)
        st.markdown(f"📊 **{top_divorced.idxmax()}** leads in the number of
divorced individuals.")

```

```

fig3 = px.bar(top_divorced, x=top_divorced.index,
y=top_divorced.values,
               title='Top 10 Countries with Highest Divorce Rates',
               labels={'y': 'Number of Divorced Individuals', 'x':
'Country'}, text_auto=True)
st.plotly_chart(fig3, use_container_width=True)

# 3. Age Group Analysis
st.subheader("📊 Age Group Analysis by Marital Status & Gender")
fig4 = px.box(filtered_df, x='MaritalStatus', y='AgeGroup', color='Sex',
title='Age Group vs Marital Status by Gender')
st.plotly_chart(fig4, use_container_width=True)

# 4. Timeline Analysis
st.subheader("📅 Marriage Trends over Time")
fig5 = px.histogram(filtered_df, x='Data Collection (Start Year)',
color='MaritalStatus',
                    title='Marital Status over Time', text_auto=True)
st.plotly_chart(fig5, use_container_width=True)

# ----- Clustering Analysis -----
st.subheader("📊 Clustering Analysis: Country, Age, Gender & Marital
Status")

# Encode categorical columns for clustering
cluster_df = filtered_df.copy()
encoder = LabelEncoder()

for col in ['Country', 'AgeGroup', 'Sex', 'MaritalStatus']:
    cluster_df[col] = encoder.fit_transform(cluster_df[col].astype(str))

# Check for enough data
if len(cluster_df) > 10:
    kmeans = KMeans(n_clusters=3, random_state=42, n_init=10) # ✅FIX:
n_init added
    cluster_df['Cluster'] = kmeans.fit_predict(cluster_df[['Country',
'AgeGroup', 'Sex', 'MaritalStatus']])

    fig6 = px.scatter_3d(cluster_df, x='Country', y='AgeGroup',
z='MaritalStatus', color='Cluster',
                        title='3D Cluster Analysis: Country, Age Group,
Marital Status')
    st.plotly_chart(fig6, use_container_width=True)

# ----- Footer -----
st.markdown("---")
st.markdown("📝 **Note**: Dataset from World Marriage Dataset (1970-2017).
Dashboard created for educational & analytical purposes.")
st.markdown("🎉 Project Analysis Completed - By Nadeem Ahamad")

```


Appendix D: Dashboard Screenshots



World Marriage Dataset - Global Trends & Insights (1970-2017)

Welcome to the Interactive Dashboard on Global Marriage Trends.

Here you can **explore, analyze, and discover patterns** in marriage, divorce, and other marital statuses across 232 countries over 5 decades.

 **Tip:** Use the sidebar to filter data dynamically by **Country, Age Group, Gender, Marital Status** to get personalized insights.



Key Insights from Filtered Data

Total Records

5884

Total Married

1219

Total Divorced

972

Most Common Status

Married

 **Top Country for Marriage (in filtered data):** Armenia

 Click to View Filtered Data (First 50 Rows)



Explore Data Filters

Select Countries

Afghanistan

Pakistan

Bangladesh

Bhutan

India

Select Age Groups

[35-39]

[30-34]

[25-29]

Select Gender

Man

Woman

Select Marital Status

Divorced

Married

Select Age Groups

[20-24]

[25-29]

[30-34]

[35-39]

[40-44]

[45-49]

[50-54]

[55-59]

[60-64]

[65+]

[10-14]

[15-24]

[25-39]

[40-64]

[0-14]

[65-69]

[70-74]



Explore Data Filters

Select Countries

Afghanistan ×

India ×

Pakistan ×

Bangladesh ×

Sri Lanka ×

Bhutan ×



Select Marital Status

Divorced ×

Married ×

Single ×

Widowed ×

Divorced or S... ×

Separated ×

Never married ×

Not in union ×

Not living tog... ×

Married or Li... ×

Widowed or ... ×

Living together ×

Consensual ... ×



Select Gender

Man ×

Woman ×



Top Country for Marriage (in filtered data): Pakistan

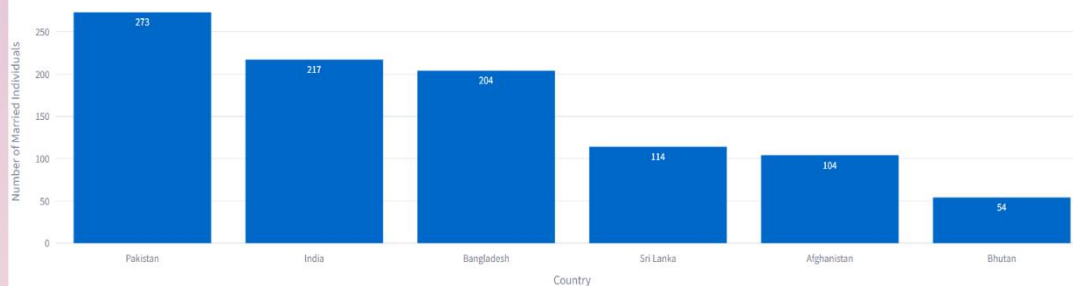
Click to View Filtered Data (First 50 Rows)

	Sr.No.	Country	AgeGroup	Sex	MaritalStatus	DataProcess	Data Collection (Start Year)	Data Collection (End Year)	Data Source
1	2	Afghanist	[20-24]	Man	Divorced	Survey	1,972	1,974	National statistics
2	3	Afghanist	[25-29]	Man	Divorced	Survey	1,972	1,974	National statistics
3	4	Afghanist	[30-34]	Man	Divorced	Survey	1,972	1,974	National statistics
4	5	Afghanist	[35-39]	Man	Divorced	Survey	1,972	1,974	National statistics
5	6	Afghanist	[40-44]	Man	Divorced	Survey	1,972	1,974	National statistics
6	7	Afghanist	[45-49]	Man	Divorced	Survey	1,972	1,974	National statistics
7	8	Afghanist	[50-54]	Man	Divorced	Survey	1,972	1,974	National statistics
8	9	Afghanist	[55-59]	Man	Divorced	Survey	1,972	1,974	National statistics
9	10	Afghanist	[60-64]	Man	Divorced	Survey	1,972	1,974	National statistics
10	11	Afghanist	[65+]	Man	Divorced	Survey	1,972	1,974	National statistics

Country-wise Marriage & Divorce Rates

✓ Pakistan has the highest number of married individuals in the selected data.

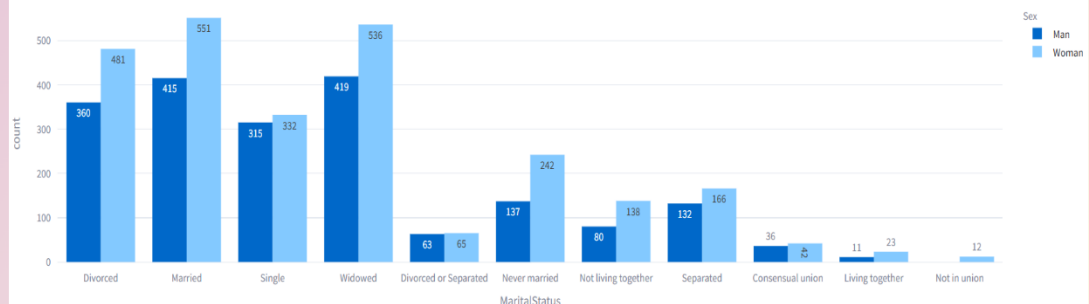
Top 10 Countries with Highest Marriage Rates

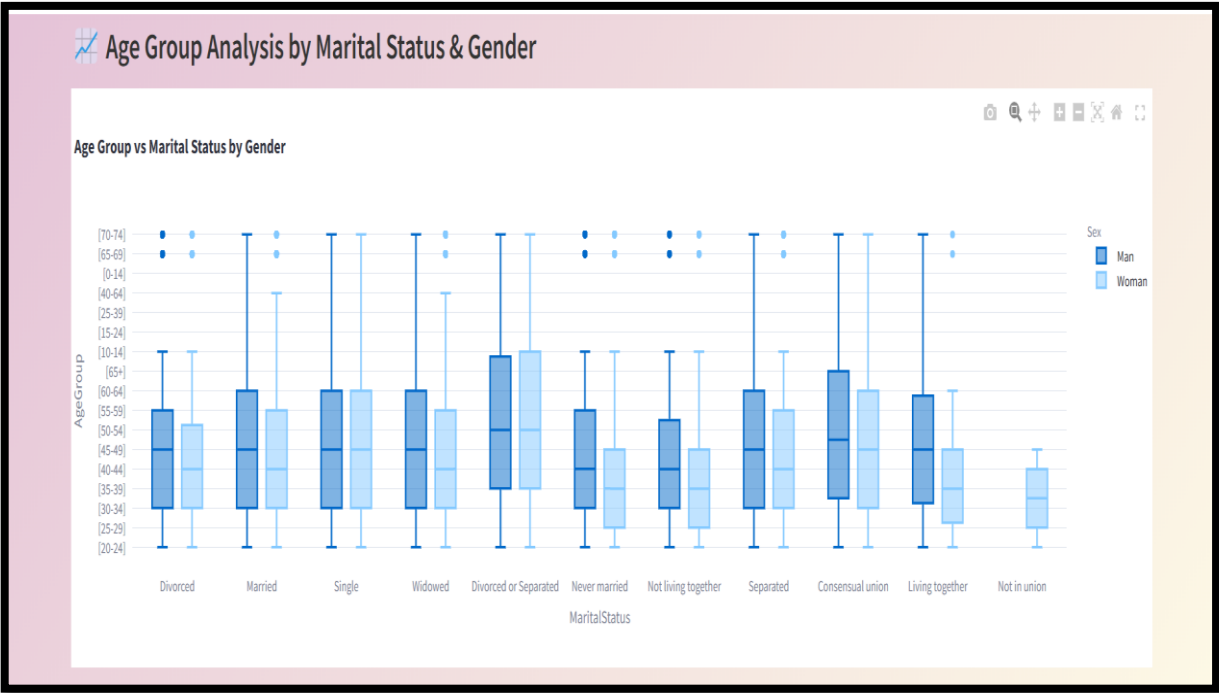
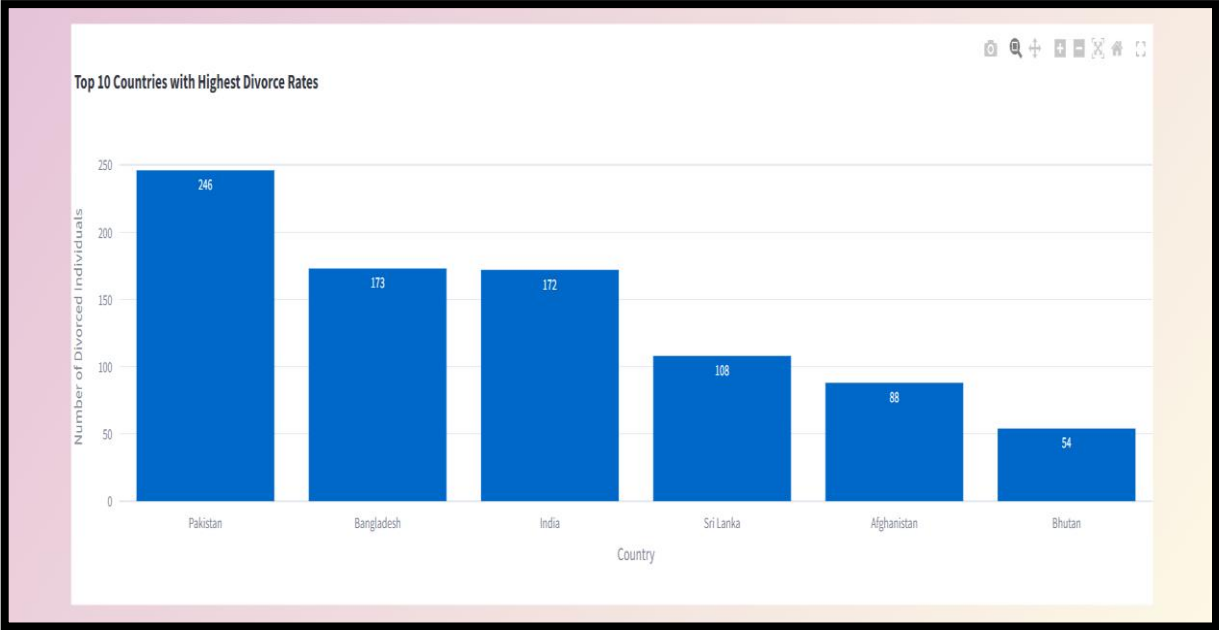


⚠ Pakistan leads in the number of divorced individuals.

Marital Status Distribution by Gender

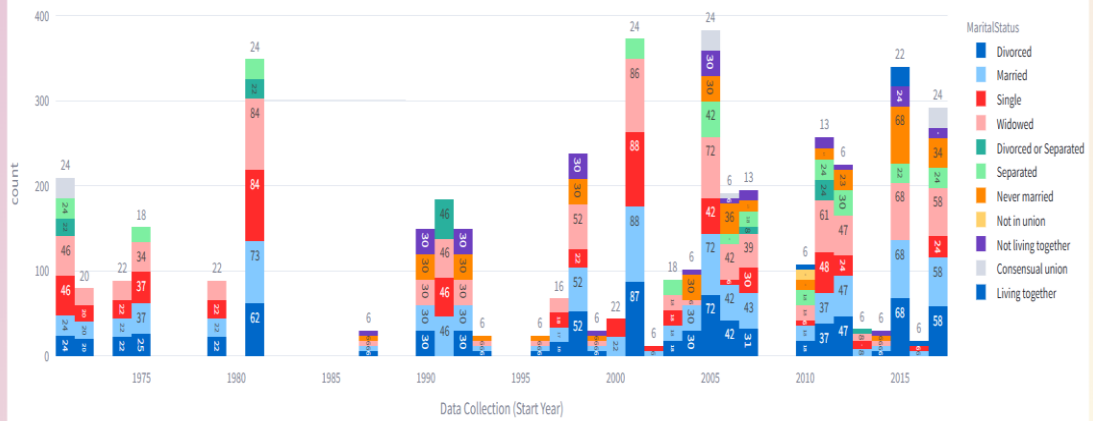
Distribution of Marital Status





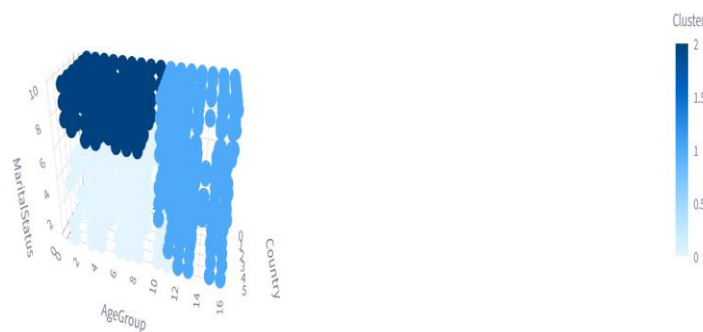
Marriage Trends over Time

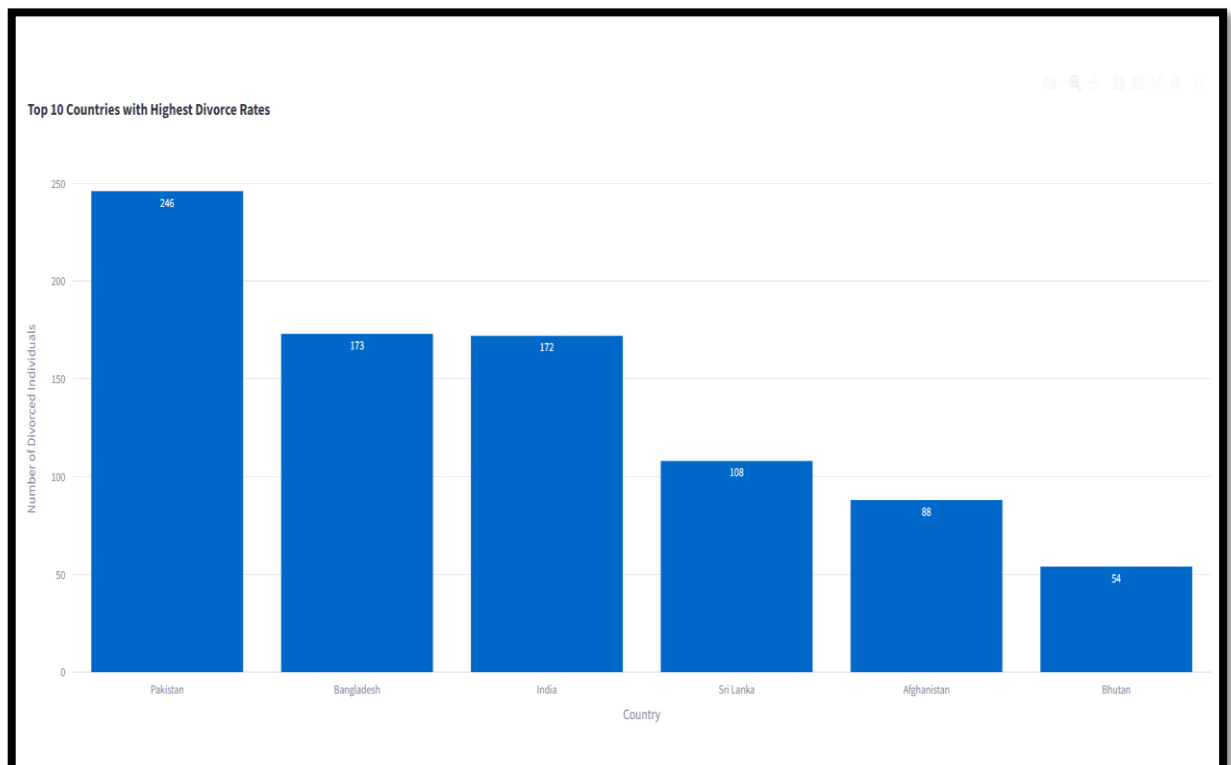
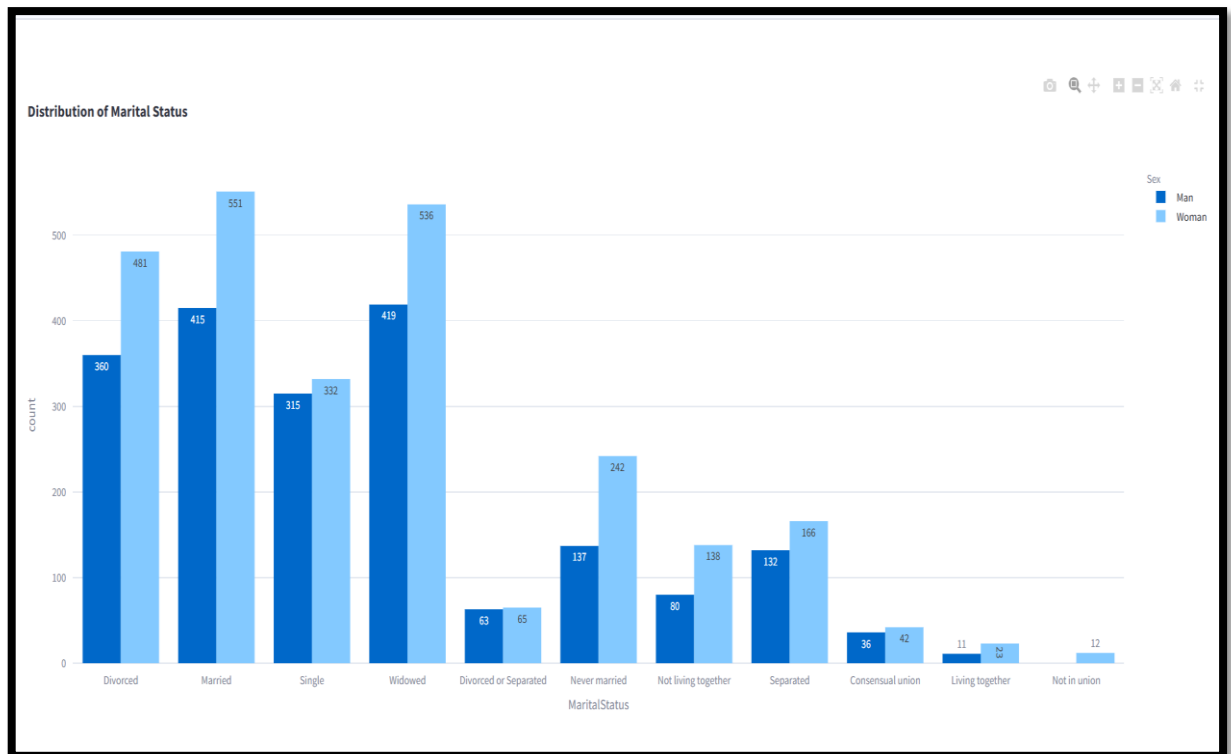
Marital Status over Time

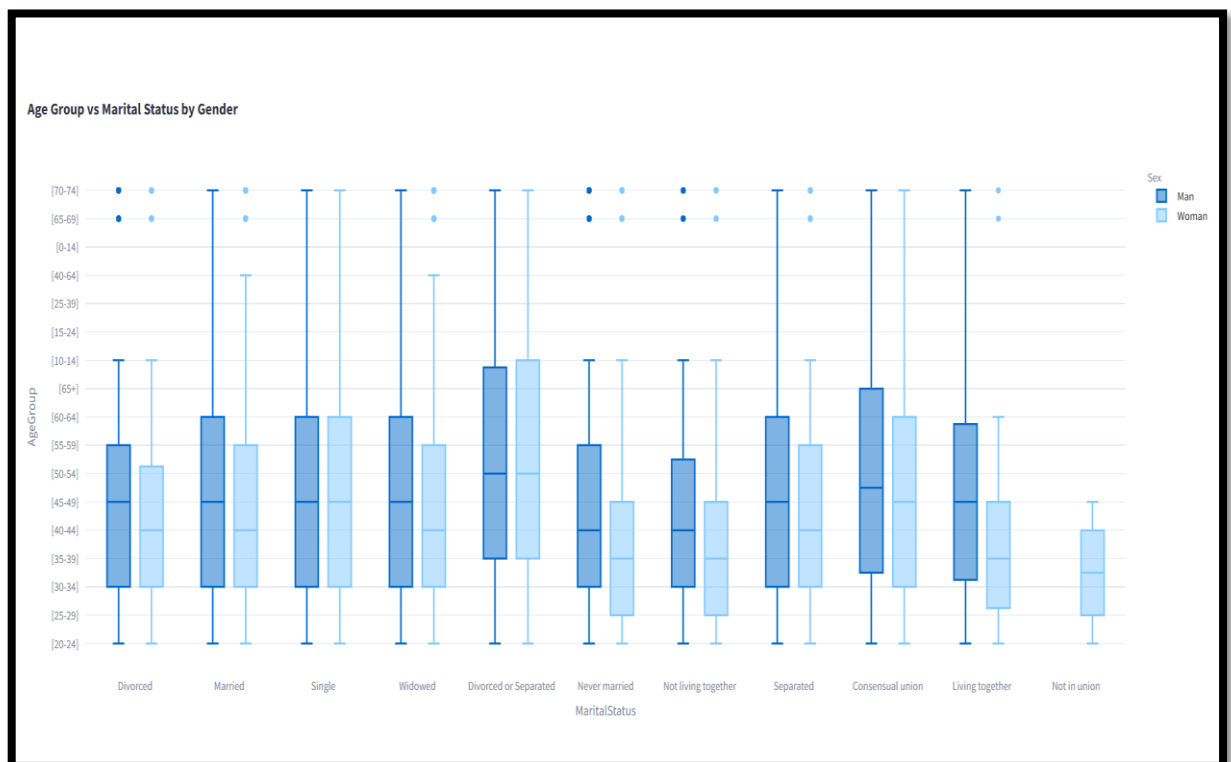
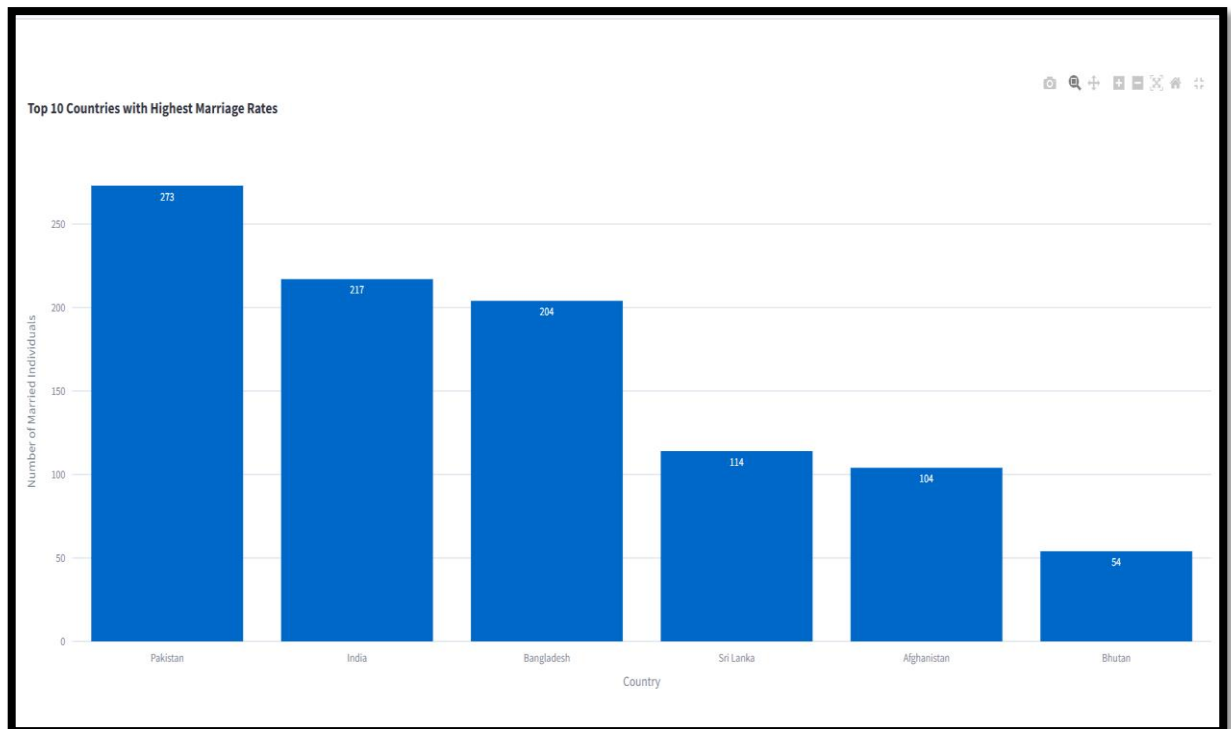


Clustering Analysis: Country, Age, Gender & Marital Status

3D Cluster Analysis: Country, Age Group, Marital Status







Clustering Analysis: Country, Age, Gender & Marital Status

3D Cluster Analysis: Country, Age Group, Marital Status



Distribution of Marital Status

